

Directional transience implies ballisticity

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Abstract

We prove that almost-sure transience in a fixed direction implies a deterministic limiting velocity with positive projection in that direction for nearest-neighbor random walks in independent and identically distributed uniformly elliptic environments on $\mathbb{Z}^d$, $d \geq 2$. This resolves the ballisticity conjecture positively.

Contents

1	Introduction	2
1.1	Model and main result	2
1.2	Context and significance	3
1.3	Proof strategy	4
1.4	Positive-probability transience	5
2	Geometric and speed reductions	5
2.1	Strict records and regeneration words	5
2.2	Finite spatial radius and the limiting ray	7
2.3	Layer notation; a small inverse moment	10
2.4	Probability estimate sufficient for speed	11
3	Two-walk scales and preliminary comparisons	13
3.1	Common true cuts	13
3.2	Truncated variance and median centering	16
3.3	Change of conditioning and short-block survival	18
3.4	Endpoint spread	20
4	Gaussian scales	21
4.1	Independent-path Brownian limit	22
4.2	Shared paths: marginals and separation	25
4.3	Zero occupation on the projected diagonal	27
4.4	Shared joint limit and quenched consequence	30
5	Clipping paths on arbitrary scales	32
5.1	Transferring truncated moments	32
5.2	A path policy on short blocks	32
6	Kernel tools	36
6.1	Creating separated seeds	36
6.2	Rapid loss bound	38
6.3	An outward-order kernel	39

7	Separated stages and episode bounds	41
7.1	Separated extensions	42
7.2	Episode operation	46
8	Stationary profiles and the contradiction	49
8.1	The stationary array	49
8.2	Entropy in a finite episode window	51
8.3	Profiles and multiplicities	52
8.4	Why a stationary profile cannot sustain normalization growth	54

1 Introduction

Directional transience says that a random walk eventually leaves every backward half-space. Ballisticity asks for more: linear progress with a strictly positive speed. For a random walk in a random environment, repeated visits to unfavorable regions can separate these two behaviors. We prove that they coincide for nearest-neighbor walks in iid uniformly elliptic environments in every dimension at least two.

1.1 Model and main result

Let $d \geq 2$, let $e_1, \dots, e_d$ be the coordinate unit vectors, and set

$$U = \{\pm e_1, \dots, \pm e_d\}, \quad \Delta_U = \left\{ p \in [0, 1]^U : \sum_{e \in U} p(e) = 1 \right\}.$$

Let ν be a probability measure on Δ_U . An environment is a collection $\omega = (\omega(x, \cdot))_{x \in \mathbb{Z}^d}$ of transition rows, with product law

$$P = \nu^{\otimes \mathbb{Z}^d}.$$

Thus different sites carry independent, identically distributed rows; no independence among the entries of one row is required. Throughout, we assume *uniform ellipticity*: there is a constant $\kappa > 0$ such that

$$\nu \left(\min_{e \in U} p(e) \geq \kappa \right) = 1.$$

For fixed ω , the quenched law $P_{x, \omega}$ is that of the Markov chain starting at x and satisfying

$$P_{x, \omega}(X_{n+1} = y + e \mid X_n = y) = \omega(y, e).$$

Its annealed law is $P_x(\cdot) = \int P_{x, \omega}(\cdot) P(d\omega)$. We also use P_x for the joint environment-path law when both coordinates are needed. Write E for environment expectation, and use the corresponding law labels on other expectations when necessary.

For a fixed $\ell \in S^{d-1}$, define

$$A_\ell = \{X_n \cdot \ell \longrightarrow +\infty\}.$$

The walk is *directionally transient* in direction ℓ if $P_0(A_\ell) = 1$, and *ballistic* in that direction if

$$P_0 \left(\liminf_{n \rightarrow \infty} \frac{X_n \cdot \ell}{n} > 0 \right) = 1.$$

Theorem 1.1 (Directional transience implies ballisticity). *Let $d \geq 2$ and let $P = \nu^{\otimes \mathbb{Z}^d}$ be an iid law of nearest-neighbor transition rows satisfying uniform ellipticity. If $\ell \in S^{d-1}$ and $P_0(A_\ell) = 1$, then there exists a deterministic vector $v = v(\nu) \in \mathbb{R}^d$ such that*

$$v \cdot \ell > 0, \quad P_0\left(\lim_{n \rightarrow \infty} \frac{X_n}{n} = v\right) = 1.$$

In particular, the walk is ballistic in the same direction ℓ .

The direction may be any fixed real unit vector. No drift, symmetry, reversibility, independence within a row, or regeneration-time moment assumption is added. The velocity is a property of the row law, since an almost-sure deterministic vector limit is unique. The assertion is for each fixed direction and does not require a common exceptional null set for all directions.

1.2 Context and significance

Theorem 1.1 resolves positively the ballisticity conjecture for iid uniformly elliptic nearest-neighbor environments in dimension at least two. Fribergh and Kious [8, Conjecture 1.1] formulate it with precisely the one-direction transience assumption used here. The distinction between escape and speed is essential: already in one dimension, uniformly elliptic iid environments can give a transient walk with zero speed; see Solomon [19] and the discussion in [21, p. 510]. The higher-dimensional conjecture asks whether the additional spatial routes prevent this form of slowdown.

The regeneration construction of Sznitman and Zerner [22] separates the walk at new height records below which it never returns. After an initial piece, the resulting pieces are independent with a common conditioned law; see also [21, Section 1] and [8, Section 2]. A finite mean duration then gives a velocity by the renewal law of large numbers. Almost-sure finiteness of the pieces does not supply that mean. The proof here first controls a piece's spatial radius—its greatest distance from its starting point—and only later its duration. Zeitouni's survey [23] gives further background on this distinction and the regeneration method.

Several quantitative routes to ballisticity precede the present result. Kalikow's auxiliary-walk approach [12] led to sufficient directional drift conditions. Under Kalikow's condition, Sznitman [20] obtained slowdown estimates and a central limit theorem through regeneration-time tails. His conditions $(T)_\gamma$ and (T') require stretched-exponential control of backward slab exits in nearby directions, and his effective criterion gives a finite-box characterization of (T') yielding positive speed [21]. Berger, Drewitz, and Ramírez [3] proved that sufficiently strong polynomial exit bounds imply (T') . Guerra and Ramírez [11] proved $(T') \Rightarrow (T)$, where (T) is the exponential slab-exit condition. Guerra's later preprint [10, Definition 1.4 and Theorem 1.9] gives a finite-box criterion already implied by backward-exit probabilities of order $o(L^{1-d})$ in nearby directions. These results start from quantitative exit assumptions. Theorem 1.1 starts with almost-sure escape in one direction and derives the quantitative crossing estimate needed for finite mean duration.

Even the spatial intermediate conclusion requires care. Simenhaus [18, Theorem 1] obtains a deterministic asymptotic direction from transience in a nonempty open set of directions. Our spatial-radius estimate uses the specified single direction and gives a limiting ray before a duration moment has been proved. The record-counting proof is related to the cone-renewal argument in [18, Lemma 2]; its treatment of arbitrary real projected heights is given directly in Section 2.

The two-walk construction has a close precedent in the common spatial regeneration levels of Rassoul-Agha and Seppäläinen [16, Section 7]. Comparing shared and independent copies

to control quenched fluctuations belongs to the second-moment approach of Bolthausen and Sznitman [5] and Berger and Zeitouni [4]. Their invariance principles assume additional ballisticity or regeneration-time moments. Here the comparisons use spatial first moments, truncated variances, and selected Gaussian scales; the quenched approximation is proved along those scales in environment probability.

Finally, normalized successful-endpoint distributions can spread out or split into groups whose mutual distances diverge. Translation-invariant compactifications retaining these possibilities were developed by Mukherjee and Varadhan [14] and, for polymer endpoints, by Bates and Chatterjee [1]. We construct the profile and upper-environment array needed here and prove its finite-window entropy bound. That bound excludes diffuse mass and infinitely many separated groups on the event of persistent survival loss. An order comparison in the two distant tails of a remaining profile then contradicts that loss.

1.3 Proof strategy

The proof separates spatial control from control of elapsed time. Section 2 constructs true record words in the original real direction. Counting ordinary record indices gives finite mean projected width, even when projected heights are not discrete. A forward template at a tilted record then gives finite mean spatial radius. This produces a deterministic limiting ray without a duration moment and permits a signed coordinate permutation so that height is x_1 .

For that coordinate height, let

$$a_N = P_{0,\omega}(T_N < T_{-1})$$

be the quenched probability of reaching level N before level -1 . Here T_j denotes the first hit of level j . The decisive estimate is (5): for a fixed $0 < \beta < 1$, the environment probability of $a_N < N^{-\beta}$ decays faster than every power of N . The same section proves that this estimate supplies the missing duration moment and hence Theorem 1.1. The rest of the paper proves the estimate.

Sections 3 and 4 compare two walks at first hits of integer layers. Independent common regeneration words give symmetric increments and a scale defined by their truncated variance. On Gaussian subsequences, a shared-environment comparison must exclude positive limiting occupation on the projected diagonal. Its cutoff-scale argument also controls the smaller scales that need not be Gaussian. The resulting two-copy limit yields a quenched approximation without assuming a second moment or a pre-existing quenched invariance principle.

Section 5 combines Gaussian and non-Gaussian subscales to retain positive quenched mass whose first-hit positions lie in narrow tubes at every large scale. Section 6 turns this into separated endpoint seeds, rapid-loss bounds, and an outward-order estimate with an exponentially integrable local remainder. All adaptive tests use only departure rows below their tested top layer.

If (5) fails, conditioning on its bad environments costs only logarithmic relative entropy. The multiscale stages of Section 7 then give a positive expected loss of global survival mass on episodes with bounded stage count. Each episode is an adaptively chosen stretch of layers; the bound on its stage count also controls the logarithm of its height. In Section 8, averaging over episodes gives a stationary law of local endpoint profiles and their upper environments. Entropy bounds exclude diffuse mass and infinitely many separated profile classes on the positive-drop event. A tail-order statistic in a surviving full-support class contradicts the sustained drop: testing far out in its two tails removes the local environmental bias from the outward-order remainder. This contradiction proves (5) and completes the speed argument.

1.4 Positive-probability transience

In dimensions at least three, a separate directional zero–one theorem allows us to assume only positive transience probability. The input is Theorem 1.1 of the companion manuscript [15]. Together with Theorem 1.1, it also identifies exactly the directions in which escape has positive probability.

Corollary 1.2 (Positive-probability transience and the velocity hemisphere). *Let $d \geq 3$ and let $P = \nu^{\otimes \mathbb{Z}^d}$ be an iid law of nearest-neighbor transition rows satisfying uniform ellipticity. For a fixed $\ell \in \mathbb{R}^d \setminus \{0\}$, set $A_\ell = \{X_n \cdot \ell \rightarrow +\infty\}$. If $P_0(A_\ell) > 0$, then there is a deterministic nonzero vector $v = v(\nu)$ such that*

$$v \cdot \ell > 0, \quad P_0\left(\lim_{n \rightarrow \infty} \frac{X_n}{n} = v\right) = 1.$$

If such a direction exists, then the deterministic sets of directions satisfy

$$\{u \in S^{d-1} : P_0(A_u) = 1\} = \{u \in S^{d-1} : P_0(A_u) > 0\} = \{u \in S^{d-1} : v \cdot u > 0\}.$$

In particular, $P_0(A_u) = 0$ for each fixed $u \in S^{d-1}$ with $v \cdot u \leq 0$, including every equator direction. The equality concerns annealed probabilities for each fixed deterministic direction, not a simultaneous pathwise assertion over all directions.

Proof. Uniform ellipticity implies the strict ellipticity assumed by the companion’s directional zero–one law. It therefore upgrades $P_0(A_\ell) > 0$ to $P_0(A_\ell) = 1$. Since $A_\ell = A_{\ell/\|\ell\|}$, Theorem 1.1 gives the stated velocity and positive projection.

Fix this vector v and a direction $u \in S^{d-1}$. If $v \cdot u > 0$, the vector law of large numbers gives $X_n \cdot u/n \rightarrow v \cdot u > 0$ almost surely, hence $P_0(A_u) = 1$. Conversely, if $P_0(A_u) > 0$, the preceding argument applied to u gives a deterministic vector w with $X_n/n \rightarrow w$ almost surely and $w \cdot u > 0$. The two probability-one convergence events intersect, and uniqueness of a vector limit gives $w = v$. Thus $v \cdot u > 0$. These two implications prove the equality of direction sets and the assertion on the closed opposite hemisphere. $\square$

Conventions. Logs are natural. Positive constants may change from line to line and may depend on the fixed row law and dimension; additional dependencies are indicated when uniformity matters. Walks in a shared environment are conditionally independent given that environment. At a finite path prefix, only rows at departure sites have contributed transition factors. This distinction is retained when a path stops on first arrival at a layer or a previously visited set.

2 Geometric and speed reductions

2.1 Strict records and regeneration words

Put $h(x) = (\ell/\|\ell\|_\infty) \cdot x$, so that a lattice step has height increment of absolute value at most one, and some step has increment exactly one. By hypothesis $h(X_n) \rightarrow +\infty$ almost surely. Let D_h be the event of never going below the starting height and put $p_h = P_0(D_h)$. An *ordinary strict record* is a first strict exceedance of the running height maximum, with time zero counted as the initial record. Such a record is *true* if the walk never subsequently goes below its height.

We will repeatedly use a basic consequence of independence over sites. The quenched weight of a finite path prefix uses only its departure rows. A continuation event supported on paths avoiding those departure sites has quenched probability measurable with respect

to rows outside that set: evaluate it using the walk killed on arrival at the forbidden set. Its probability therefore factors from the prefix weight after annealing. The same statement holds for infinite avoidance events by the cylinder construction of path measures. For a finite continuation, it suffices that its departure sites be outside the old departure set. None of these facts requires independence of the entries within a row.

This construction belongs to the regeneration method of Sznitman and Zerner [22]; see also [21, Section 1]. We give its needed details, including arbitrary real height, directly. The projected-width counting argument is related to the cone-regeneration renewal argument in Simenhaus [18, Lemma 2], who credits Zerner for that renewal argument. Here we count ordinary record indices so that the projected heights themselves need not form a lattice.

Lemma 2.1 (True words and their widths). *We have $p_h > 0$. Under $P_0(\cdot \mid D_h)$, include time zero as a true boundary. The relative finite path words between consecutive true boundaries are iid. The raw path has a finite first positive-time true record, after which its translated suffix has this conditioned law.*

For a conditioned word, let $L > 0$ be its height gain, let R be its maximal Euclidean distance from its starting point, including its endpoint, and let S_ count its ordinary strict records after its start and through its end. Then*

$$0 < \mathbb{E}L \leq \mathbb{E}S_* \leq p_h^{-1}.$$

Here and until the change to coordinate height below, word expectations refer to this conditioned word law.

Proof. The global height minimum is attained almost surely. Thus for some deterministic n, x the event that $X_n = x$ and that all later heights are at least $h(x)$ has positive probability. The quenched Markov property at n gives

$$E[P_{0,\omega}(X_n = x)P_{x,\omega}(D_h)] > 0.$$

Since the first factor is at most one, translation invariance gives $p_h > 0$.

At a first strict record, every earlier departure row has smaller height. The fresh-row rule therefore gives conditional raw probability p_h that the record is true. Test the first positive-time strict record. If an actual drop later detects its failure, test the next strict record above the whole past observed by that failure time. All candidates and failure detections are path stopping rules. Transience supplies another candidate after every finite failure, and each candidate has success probability p_h . Thus a true record is eventually found almost surely.

For the iid assertion, enumerate the possible first words. A word from 0 to z has positive length, every preterminal height in $[0, h(z))$, and every intermediate strict record invalidated by a drop before its end. Together with a suffix staying at or above $h(z)$, these conditions say exactly that this is the first word under D_h . Indeed a later suffix above $h(z)$ cannot invalidate any earlier lower record. The prefix weight and the suffix factor p_h use disjoint departure rows. Dividing by the initial conditioning probability p_h leaves the raw prefix weight as the word probability and an independent translated suffix of the original conditioned law. Countable enumeration and iteration give the iid words and infinitely many true boundaries. For the first raw true record the same argument applies, without the restriction that its prefix have nonnegative height. This argument does not apply an annealed Markov property at a future-dependent cut.

Under D_h , a fixed positive ordinary record index i is true with probability equal to the raw probability of reaching that record before dropping below zero: the fresh suffix contributes p_h , which cancels the conditioning. This probability is at least p_h , since all record indices are reached on D_h . The true indices are a renewal process with iid increments S_* . Its

positive-index count through m , divided by m , converges almost surely and in expectation to $1/\mathbb{E}S_*$. This remains true with limit zero when the mean is infinite, by the strong law on truncations; the ratios are bounded by one. The preceding lower bound on each indicator therefore yields $\mathbb{E}S_* \leq p_h^{-1}$. Each ordinary record increases height by at most one, so $L \leq S_*$. This argument uses record indices, not a lattice structure for the values of h . $\square$

2.2 Finite spatial radius and the limiting ray

The finite mean width controls progress only at true boundaries. To control the path between them, we prove that the spatial radius of a word also has finite mean. The argument compares two bounds for a large displacement relative to the running height maximum. An infinite mean radius would make such a displacement occur with probability of order $1/B$. At each new ratio threshold, however, there is a fixed positive chance of a fresh continuation that caps the ratio until the walk leaves the height strip. Iterating these chances gives an exponentially small upper bound in B .

Proposition 2.2 (Integrable spatial radius). *Under the conditioned word law,*

$$\mathbb{E}R < \infty. \quad (1)$$

No moment of the word duration is assumed.

Proof. Suppose instead that $\mathbb{E}R = \infty$, and put $I(s) = \mathbb{E} \min(R, s)$. This function is increasing, concave, unbounded, and satisfies $I(s) = o(s)$, since R is finite almost surely. Write $H_k = \sum_{j=1}^k L_j$, with $H_0 = 0$.

A family of likely finite templates. The width strong law gives constants $c > 0$, C , and C_0 such that

$$ck - C_0 \leq H_k \leq Ck + C_0 \quad (k \geq 0)$$

holds with probability greater than .95. Fix an integer C_1 with $C_1c > C + 4$, then $b > 1/c$, and then $\eta > 0$ so small that $4C_1\eta < .1$. For large a , choose the least positive integer N with $I(aN) \geq \eta a$. Then $N \rightarrow \infty$, and minimality and concavity give

$$\eta a \leq I(aN) < 2\eta a$$

for all sufficiently large a .

With probability greater than .8, the first C_1N words satisfy the height bounds above and

$$\sum_{j=1}^k R_j \leq ak \quad (1 \leq k \leq C_1N). \quad (2)$$

Here is the maximal estimate behind this assertion. Truncate each radius at aC_1N . Its expectation is at most $C_1I(aN) < 2C_1\eta a$. The union bound for a truncation among the first C_1N radii is therefore at most $2C_1\eta$. For a stationary nonnegative sequence (Z_j) ,

$$\Pr \left(\max_{1 \leq k \leq n} \frac{1}{k} \sum_{j=1}^k Z_j > a \right) \leq \frac{\mathbb{E}Z_1}{a}.$$

To see this, mark the starts among $1, \dots, M$ admitting a witness interval of length at most n with average greater than a . Select such intervals greedily from left to right. They are disjoint and cover all marked starts, so their total sum is greater than a times the number of marked starts. The selected intervals lie in $1, \dots, M + n$. Take expectations, divide by M , and let $M \rightarrow \infty$ to remove the boundary contribution. Applied to the truncated radii, this

costs at most another $2C_1\eta$. Together with the height event this proves the asserted .8 bound. This is the nonnegative-sequence form of Hopf's maximal ergodic inequality; see Garsia [9].

The polynomial lower bound. Fix a large $B > 1$. Choose i_0 so that

$$\sum_{i \geq i_0} \Pr(L > i) < \frac{\eta}{4B}, \quad C(i-1) + C_0 + i \leq (C+2)i \quad (i \geq i_0).$$

Choose a sufficiently large that $N \geq i_0$ and $I(Bai_0) < \eta a/4$. Tail integration, followed by removal of the width exceptions, gives

$$\frac{\eta}{2B} \leq \lambda := \sum_{i=i_0}^N \Pr(R > Bai, L \leq i) \leq 2\eta.$$

For the lower bound, the unrestricted sum is at least $[I(Ba(N+1)) - I(Bai_0)]/(Ba) \geq 3\eta/(4B)$; for the upper bound it is at most $I(BaN)/(Ba) \leq I(aN)/a$.

Count the indices i at which this large-word event occurs and the preceding words satisfy the height and radius bounds through $i-1$. Independence gives mean at least $.8\lambda$. Dropping the preceding-word restrictions bounds the second moment by $\lambda + \lambda^2$. The count is therefore positive with probability at least c_2/B , where $c_2 > 0$ does not depend on B or a . At some point of that word,

$$\|X_n\| > (B-1)ai, \quad 0 \leq h(X_j), \quad M_n := \max_{j \leq n} h(X_j) \leq (C+2)i.$$

The height bound holds throughout this prefix because every word ends at a strict record and has no earlier height above its endpoint. One of the $2d$ signed coordinate vectors u thus satisfies

$$\frac{Y(X_n)}{a(1+M_n)} > c_3B, \quad Y(x) = u \cdot x + bah(x), \quad (3)$$

for a fixed $c_3 > 0$ and sufficiently large fixed B . Consequently, under the raw law the union of these events over u has probability at least $p_h c_2/B$, before the path leaves $\{h \geq 0, M \leq (C+2)N\}$.

A fresh continuation caps the ratio. Fix u and put $Q_n = Y(X_n)/[a(1+M_n)]$. Let ζ be the exit time just described. We claim that constants $\epsilon_0 > 0$ and $K < \infty$, independent of B, a and the threshold $t > 0$, have the following property: at every finite prefix ending at the first crossing $\sigma < \zeta$ of $Q > t$, a continuation of conditional raw probability at least ϵ_0 exits the strip before Q exceeds $t+K$.

At the crossing,

$$Y(X_j) \leq ta(1+M_j) \leq ta(1+M_\sigma) < Y(X_\sigma) \quad (j < \sigma).$$

Thus the arrival is a strict Y -record. A step changes Y by at most $ba+1$, so $Q_\sigma \leq t+b+1$ when $a \geq 1$. Choose an integer $M'_0 > C_0$ with $M'_0 b > 1 + bC_0$. First prescribe M'_0 steps of height $+1$. Each increases Y by at least $ba-1 > 0$. At their endpoint impose a translated D_h -suffix whose first C_1N words satisfy the template above. In its word $k+1$, every site has Y -increment from the launch at least

$$M'_0(ba-1) + ba(ck - C_0) - a(k+1) > 0.$$

This holds uniformly in $0 \leq k < C_1N$ for large a : the coefficient of k is $a(bc-1) > 0$, and the constant term is positive by the choice of M'_0 . These words therefore avoid every old departure row. Their terminal true cut has height at least

$$h(X_\sigma) + M'_0 + cC_1N - C_0 > (C+2)N.$$

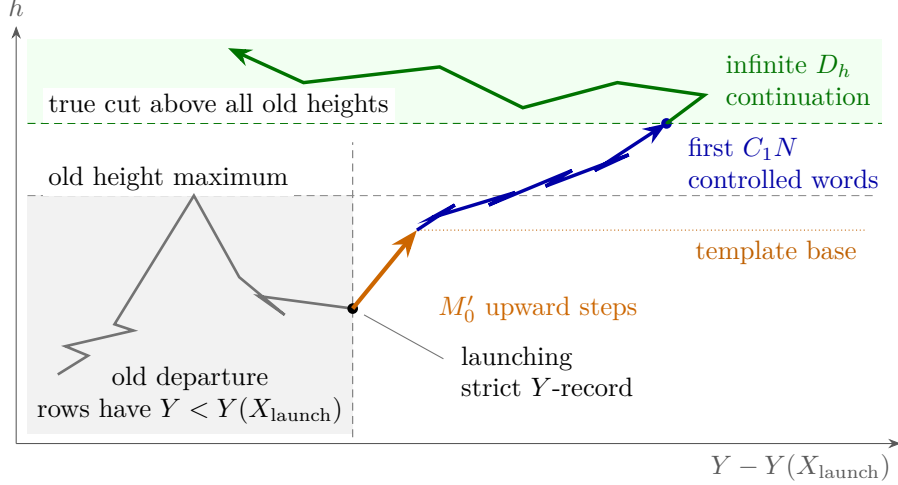


Figure 1: The forward continuation in the (Y, h) projection (schematic, not to scale). Before launch, all departure rows have smaller Y . The upward script raises the template base. The controlled words stay beyond the old Y maximum and above the script departures. Their terminal true cut lies above all old heights; after that cut the suffix may cross the old Y barrier while still avoiding the old rows. The projection does not restrict the walk to two dimensions.

The entire remaining suffix stays above this cut, and so also avoids all old departures. The template stays above its own starting height, which is strictly above every script departure. Thus the whole infinite continuation uses no old departure row: first the Y barrier provides separation, and then the terminal height does. The fresh-row factorization gives probability at least

$$\epsilon_0 = .8 \kappa^{M'_0} p_h > 0.$$

Figure 1 records these two separation mechanisms.

It remains to bound the ratio on this continuation. The positive part of the launch coordinate $u \cdot X_\sigma$, divided by any later denominator, is at most $t + b + 1$, since $h(X_\sigma) \geq 0$ and the denominator is nondecreasing. The later height term contributes at most b . The script adds at most M'_0 to the ratio. In word $k + 1$, the additional coordinate contribution is at most

$$\frac{M'_0/a + k + 1}{A_0 + ck} \leq \max \left\{ \frac{M'_0 + 1}{A_0}, \frac{1}{c} \right\}, \quad A_0 = 1 + M'_0 - C_0 > 1.$$

The numerator uses the bound on the sum of radii through word $k + 1$; the denominator uses the height of that word's starting boundary. Thus any fixed

$$K > 2b + 1 + \max \left\{ M'_0, \frac{M'_0 + 1}{A_0}, \frac{1}{c} \right\}$$

proves the claim. Exit occurs by the last prescribed template endpoint.

The exponential upper bound. Let σ_j be the first crossing before ζ of $t_j = 1 + jK$. The continuation just constructed prevents $\sigma_{j+1} < \zeta$. Summing its bound over the finite prefixes at σ_j yields

$$P_0(\sigma_{j+1} < \zeta) \leq (1 - \epsilon_0)P_0(\sigma_j < \zeta).$$

This conditions only on the observed crossing prefix; it does not reveal whether an infinite template has failed. Hence the event in (3) has probability at most $C'e^{-c'B}$ for this u . Taking the union over signed coordinates contradicts the lower bound $p_h c_2/B$ when B is large. All

estimates use the permitted order: fix the geometry constants, then B , then i_0 , then take a large. This proves (1). $\square$

Corollary 2.3 (Deterministic limiting ray). *There is a deterministic unit vector r with $h(r) > 0$ such that $X_n/\|X_n\| \rightarrow r$ almost surely. After a deterministic signed permutation of the coordinate axes, $r_1 > 0$, and all the preceding word conclusions hold with height x_1 .*

Proof. Word endpoints obey the vector strong law, with mean displacement w satisfying $h(w) = \mathbb{E}L > 0$. Proposition 2.2 also gives $R_k/k \rightarrow 0$ almost surely: for each $\varepsilon > 0$, the probabilities $\Pr(R_k > \varepsilon k)$ are summable. Thus interpolation inside each word gives the same limiting direction $r = w/\|w\|$, without any duration moment. The completed-word count tends to infinity. Lemma 2.1 transfers the conclusion through the finite initial raw prefix.

Some signed coordinate of r is positive. Relabel it as the first coordinate, preserving the lattice, uniform ellipticity, and the iid row law. Then $(X_n)_1 \rightarrow +\infty$ almost surely. The proofs of Lemma 2.1 and Proposition 2.2 apply again with this coordinate height, using only its transience. We retain the same notation for the transformed walks and laws. $\square$

2.3 Layer notation; a small inverse moment

Henceforth height means x_1 . Write $x = (x_1, \underline{x})$, $\underline{x} \in \mathbb{Z}^{d-1}$; for fluctuations we use the single scalar coordinate x_2 . For a path from x and $j \in \mathbb{Z}$, $T_j = \inf\{n \geq 0 : (X_n)_1 = x_1 + j\}$ denotes first hit of the layer $x_1 + j$, relative to the indicated start ($\inf \emptyset = \infty$; $T_0 = 0$). Put $D = \{T_{-1} = \infty\}$, $p = P_x(D) > 0$,

$$P_x^+ = P_x(\cdot \mid D), \quad q(x) = P_{x,\omega}(D), \quad a_H(x) = P_{x,\omega}(T_H < T_{-1}) \quad (H \geq 0 \text{ integer}),$$

where the environment in notation such as q, a is implicit. Let $K_H(x, \cdot)$ be the unnormalized quenched kernel of reaching that upper layer before the drop, of mass $a_H(x)$; we also use it for the whole path prefix through this first hit, testing events on that prefix. It only uses rows in the H layers from the starting layer inclusive to the upper one exclusive. From 0 abbreviate $a_H = a_H(0)$, $u_H = Ea_H$. The filtration $\mathcal{F}_H$ on environments reveals rows at $0 \leq x_1 < H$; under P the upper layers from H inclusive are thus fresh iid layers, also starting at any finite $\mathcal{F}_H$ -stopping layer (by partitioning according to that layer). Corresponding shifted filtrations at other bases will be used similarly. Write L_s, R_s, τ_s for the height width, radius and number of path steps in a single-walk true word in direction e_1 under P_0^+ .

Lemma 2.4 (A small inverse moment). *For every site x , $a_H(x) \downarrow q(x) > 0$ for almost every environment. Moreover, for some $\lambda > 0$,*

$$E q(0)^{-\lambda} < \infty. \tag{4}$$

Proof. Uniform ellipticity excludes confinement forever in a fixed finite-height strip, even quenched: blocks of sufficiently many consecutive upward steps have a fixed positive chance. Hence $a_H(x) \downarrow q(x)$. Also $q(je_1)$ uses only layers at or above j , so the event that $q(je_1) > 0$ infinitely often is a tail event of the independent layers. For every m , the union of these events over $j \geq m$ has probability at least $P(q(0) > 0) > 0$, by stationarity and $Eq(0) = p > 0$. The tail zero-one law makes their limsup almost sure. A finite upward script connects zero to one such site and gives $q(0) > 0$. Translation and countability give positivity at every site simultaneously.

At layer H , let $\mu_H = K_H(0, \cdot)/a_H$ be the endpoint probability distribution. It is $\mathcal{F}_H$ -measurable, and the quenched Markov property, restricted to a suffix never below layer H , gives

$$q(0) \geq a_H \int q(x) \mu_H(dx).$$

Each $q(x)$ on that layer has the original marginal law independently of the lower rows. If $q(0)/a_H < s$, then more than half the μ_H -mass has $q(x) \leq 2s$. Conditional expectation followed by Markov's inequality therefore gives

$$P(q(0)/a_H < s \mid \mathcal{F}_H) \leq 2P(q(0) \leq 2s).$$

The endpoint values $q(x)$ need not be independent of one another. Partitioning by a finite stopping layer gives the same conditional bound there.

Choose $0 < \delta < \kappa$ sufficiently small that $\rho := 2P(q(0) \leq 2\delta/\kappa) < 1$, and set

$$H_j = \inf\{H \geq 0 : a_H < \delta^j\}, \quad j \geq 1.$$

These are stopping layers. One upward step after a successful crossing gives $a_{H+1} \geq \kappa a_H$, so on $H_j < \infty$, $a_{H_j} \geq \kappa \delta^j$. The event $H_{j+1} < \infty$ implies $q(0) < \delta^{j+1}$, and hence $q(0)/a_{H_j} < \delta/\kappa$. The preceding conditional bound yields

$$P(H_{j+1} < \infty \mid \mathcal{F}_{H_j}) \leq \rho \quad \text{on } \{H_j < \infty\}.$$

Since $a_H \downarrow q(0)$, the event $H_j < \infty$ is exactly $\{q(0) < \delta^j\}$. Iteration gives its probability at most ρ^{j-1} . Summing over $\delta^{j+1} \leq q(0) < \delta^j$, and choosing $\lambda > 0$ with $\rho\delta^{-\lambda} < 1$, proves (4). $\square$

2.4 Probability estimate sufficient for speed

The rest of the paper will establish the following estimate for one fixed $0 < \beta < 1$:

$$P(a_N < N^{-\beta}) < N^{-D_0} \quad \text{eventually for each fixed finite } D_0 > 0. \quad (5)$$

Small quenched exit probabilities also connect spatial estimates to regeneration-time tails in Sznitman's work [21, Lemma 3.2 and Theorem 3.3]. The next proposition gives the reduction for the crossing probabilities defined here, with (5) as its explicit input.

Proposition 2.5 (From crossing probabilities to speed). *If (5) holds, then $\mathbb{E}\tau_s < \infty$, and the walk has an almost-sure deterministic velocity with strictly positive projection on the original direction ℓ .*

Proof. All conclusions below using (5) are conditional on that estimate.

Spatial confinement before exit. The raw path before leaving the height interval $[0, N]$ stays in $\|\underline{X}\|_\infty \leq CN^2$, except with probability at most Ce^{-cN} . To prove this, fix a signed lateral coordinate u . By Corollary 2.3, a sufficiently large fixed $b > 0$ makes $Y(x) = u \cdot x + bx_1$ transient in its positive direction. Thus the event of staying forever at $Y \geq 0$ has positive raw probability, by the minimum argument of Lemma 2.1.

For a sufficiently large fixed C_2 , the path through its first relative height hit $N + 1$ has norm at most C_2N with probability tending to one. Indeed the ray gives $\|X_n\| \leq (C_2/2)(X_n)_1$ eventually, and the earlier path has a finite maximum norm. The intersection of this event with the preceding no-backtracking event therefore has probability bounded below uniformly for large N .

At a first crossing of $Y > jC_3N$, before strip exit, the walk is at a strict Y -record. Impose that favorable translated suffix using the fresh rows at or above the record. If C_3 is sufficiently large, the suffix precludes crossing $(j+1)C_3N$ before strip exit: its Y -displacement through relative height $N + 1$ is at most $(1+b)C_2N$, the launch overshoot is bounded, and absolute strip exit occurs no later than this relative height hit. The finite-prefix threshold recursion used in Proposition 2.2, now with spacing C_3N , gives exponential decay through

N thresholds. Since $u \cdot X \leq Y$ before exit, the finite union over signed lateral coordinates proves the confinement assertion.

Super-polynomial width tails.

Now $\Delta u_N := u_N - u_{N+1} \geq 0$ is the annealed probability of reaching N before the drop but dropping before $N + 1$. In environments good at all sites of the strip and box (where $a_{N+1}(x) \geq (N + 1)^{-\beta}$), the portion of this event inside the box has quenched probability at most

$$(1 - (N + 1)^{-\beta})^{N+1}.$$

Indeed, after reaching N , at first descending-level visits to each of $N, N - 1, \dots, 0$ still in the box, there is at least the indicated chance to hit top $N + 1$ before the next descent by one. Condition (5), translation, a polynomial union bound, and the box claim thus give super-polynomial decay of Δu_N .

The true levels under P_0^+ form the renewal process of iid widths L_s , with renewal masses exactly u_n : impose the fresh D -suffix at the first reach of n before drop, and cancel p . If $G_j = P_0^+(L_s > j)$, then $\sum_{j=0}^n G_j u_{n-j} = 1$ by the last renewal up to n , so

$$G_{n+1} = \sum_{j=0}^n G_j \Delta u_{n-j}.$$

Put $f_k = \Delta u_{k-1}$ for $k \geq 1$. This nonnegative kernel has total mass $\theta = 1 - p < 1$ and all polynomial moments. The recurrence, with $G_0 = 1$, gives $G = \sum_{r \geq 0} f^{*r}$, including unit mass at zero for $r = 0$. If $\theta > 0$, normalize f by θ . For every positive integer m , the m -th moment of its r -fold convolution is at most r^m times its m -th moment, by $(z_1 + \dots + z_r)^m \leq r^{m-1} \sum_i z_i^m$. Summing with the geometric weights θ^r proves $\sum_n n^m G_n < \infty$. The case $\theta = 0$ is immediate. In particular the width tail G_n decays super-polynomially.

Time integrability and velocity. The raw exit time $T_N \wedge T_{-1}$ exceeds N^A with super-polynomially small probability, for a sufficiently large fixed A . To see this, use confinement and (5) on a slightly enlarged box and strip, requiring $a_N(x + e_1) \geq N^{-\beta}$ for every relevant site with $0 \leq x_1 < N$. A polynomial union bound controls the exceptional environments.

From any visit to such a site x , the quenched probability of no further visit before exit is at least $\kappa a_N(x + e_1)$: take one upward step and then advance N before dropping below the new layer. This event forces original strip exit before revisiting x , even though the auxiliary target may be higher than N . The quenched Markov property consequently bounds each site's visit count by a geometric tail with success probability at least $\kappa N^{-\beta}$. There are at most CN^{2d-1} sites in the confined box. If the exit time exceeds N^A , one site has at least $N^A/(CN^{2d-1})$ visits. For $A > 2d - 1 + \beta$, the union of these geometric-tail events is super-polynomially small, as are the bad-box and bad-environment events.

On D , if $L_s \leq N$, the first word ends at its first arrival to height L_s , and hence $\tau_s \leq T_N$. Thus

$$P_0^+(\tau_s > N^A) \leq P_0^+(L_s > N) + p^{-1} P_0(T_N \wedge T_{-1} > N^A).$$

Both terms have super-polynomial decay, proving $\mathbb{E}\tau_s < \infty$. The iid-word strong laws now give the velocity as mean displacement divided by mean duration; its first coordinate is $\mathbb{E}L_s/\mathbb{E}\tau_s > 0$. The integrable spatial radius controls the inter-endpoint deviations, so this limit holds at every path time. The finite raw initial prefix does not change it. Finally the velocity is a positive multiple of the ray in Corollary 2.3, which has positive projection on the original ℓ . Undoing the signed coordinate permutation proves the stated conclusion. As an almost-sure deterministic limit, this velocity is unique for the law ν . $\square$

It remains to prove (5). Sections 3–8 are devoted to that estimate; no finite duration moment is used there.

3 Two-walk scales and preliminary comparisons

We now measure fluctuations in the second coordinate at first hits of successive height layers. Common true levels provide fresh pair continuations and, for independent environments, an iid sequence of symmetric increments. Their truncated variance will determine the fluctuation scale. We first compare all first-hit positions with this sequence, then transfer the comparison between two forms of conditioning. The resulting short-block estimate and endpoint spread bounds will be used in Sections 4 and 5.

On a successful prefix, set

$$Y_j = (X_{T_j} - X_0)_2.$$

Choose a deterministic integer median b_j of Y_j under P_0^+ , with $b_0 = 0$, and write

$$W_H = \max_{0 \leq j \leq H} |Y_j - b_j|.$$

All these variables record displacement from the indicated starting site.

3.1 Common true cuts

Common spatial regeneration levels for two walks, reached at possibly different path times, are central to the shared- and independent-environment comparison of Rassoul-Agha and Seppäläinen [16, Section 7]. Their invariance principle assumes regeneration-time moments. Here we establish the common-cut identities directly from finite words, using the spatial first moment proved in Section 2.

For two starting sites x, y on the same layer, we distinguish the independent conditioned law $P_x^+ \otimes P_y^+$ from the annealed law $P_{x,y}^{++}$ of two walks in the same environment, conditioned on both no-drop events D . Let $P_{x,y}^{\text{sh}}$ denote the latter pair law before conditioning, and put

$$p_2(x, y) = E[q(x)q(y)].$$

The positivity of q established in Section 2 gives a uniform lower bound

$$\underline{p}_2 := \inf_{x_1=y_1} p_2(x, y) > 0.$$

Indeed, for every $c > 0$,

$$P(\min(q(x), q(y)) < c) \leq 2P(q(0) < c),$$

so a sufficiently small c makes $q(x)q(y) \geq c^2$ on an event of uniformly positive probability. Product-field mixing also gives

$$p_2(x, y) \longrightarrow p^2 \quad \text{as } \|x - y\| \longrightarrow \infty, \quad x_1 = y_1.$$

To verify this directly, approximate $q(0)$ in $L^1(P)$ by a bounded function of finitely many rows and translate the approximation to x and y . The approximating functions are independent once their translated row sets are disjoint.

Each marginal of the shared conditioned law is dominated uniformly by the corresponding single conditioned law:

$$P_{x,y}^{++}(X^{(1)} \in A) = \frac{E[P_{x,\omega}(A \cap D)q(y)]}{p_2(x, y)} \leq \frac{p}{\underline{p}_2} P_x^+(A),$$

and likewise for the second walk. With superscripts distinguishing the walks, define

$$\Delta_j = Y_j^{(2)} - Y_j^{(1)}, \quad Z_j = y_2 - x_2 + \Delta_j.$$

A *common true level* is a level that is true for both walks at their respective first hits. Heights are measured from the common starting layer, and level zero is included under either conditioned law.

Lemma 3.1 (Common words and their continuation laws). *Under either conditioned pair law, common true levels occur infinitely often. At each such level the translated pair continuation is fresh: under independent environments it has law $P_0^+ \otimes P_0^+$, while under a shared environment, from terminal sites x', y' , it has law $P_{x', y'}^{++}$ before translation. The latter transition law depends on the terminal pair of sites.*

These continuation rules remain valid at finite stopping indices of the chain of common words, when the information retained consists of the traversed words and their path prefixes.

Proof. Work first with the raw pair laws. Test a new integer level above both explored path pasts, pausing each walk at its first arrival there. All departure rows revealed by these prefixes lie strictly below the candidate level. The conditional probability that both arrivals are true is p^2 in independent environments and p_2 at the arrival sites in a shared environment. Both probabilities have a uniform positive lower bound.

If the candidate fails, resume the two walks, for example one step of each per round, until one drops below the candidate. This failure is detected in finite time. Then test a level strictly above both explored pasts. Each walk tends to positive infinite height almost surely; this also holds for the shared raw pair by its marginals. Thus, after every detected failure, another candidate can be reached. The uniform success bound proves the existence of a common true level above any prescribed height. Conditioning on the two initial no-drop events preserves this almost-sure conclusion.

For the continuation law, specify a pair of finite words from x, y to their first hits x', y' of a common terminal level $L > 0$. Call this word pair admissible when both prefixes avoid the initial drop and every intermediate positive level has a witnessed failure of common truth: at least one prefix has dropped below that level after its first hit. These are exactly the required prefix conditions. Indeed, once truth is imposed at L , any failure of truth at an earlier level must already have occurred in the specified prefixes.

Let $A_{x, y}^{\text{sh}}(w_1, w_2)$ be the raw annealed prefix weight of an admissible word pair. Its quenched weight is the product of the two path weights in the shared environment, including any repeated use of a row. Every departure row lies below the terminal layer. The suffixes constrained by truth use only rows at or above that layer, so site-row independence gives the conditional word probability

$$P_{x, y}^{++}((w_1, w_2) \text{ is the first common word}) = A_{x, y}^{\text{sh}}(w_1, w_2) \frac{p_2(x', y')}{p_2(x, y)}.$$

More generally, for any event B of the suffix pair, its joint probability with this word pair is

$$A_{x, y}^{\text{sh}}(w_1, w_2) \frac{p_2(x', y')}{p_2(x, y)} P_{x', y'}^{++}(B).$$

Here B is read with the new sites as starting sites. In particular, the factor $p_2(x', y')/p_2(x, y)$ is part of the shared word law; it does not cancel in general.

In independent environments the corresponding terminal and initial conditioning probabilities are both p^2 , so their ratio is one. Translation of each walk separately then makes the relative suffix law $P_0^+ \otimes P_0^+$, independent of the word just traversed. Enumerate the countably many admissible finite word pairs and iterate these factorizations to obtain the stated continuation rules. Summing over a finite stopping index gives the same rules there. The information at such an index does not include any unseen upper environment. $\square$

In independent environments, let $L \geq 1$ be the integer width of a common word and let S be its increment of Δ . The successive pairs (L_i, S_i) , with their path decorations, are iid by Lemma 3.1. Exchanging the walks shows that S is symmetric. In what follows, expectations of L, S refer to this independent common-word law.

Lemma 3.2 (Common-word moments and local tightness). *The independent common-word variables satisfy*

$$1 \leq \bar{L} := \mathbb{E}L \leq p^{-2}, \quad \mathbb{E}|S| < \infty, \quad \Pr(S \neq 0) > 0.$$

For a single conditioned walk, and for either marginal of a shared conditioned pair, the increments $Y_{h+j} - Y_h$ are tight uniformly in the deterministic level h when j ranges over any fixed bounded set of nonnegative integers.

If $\sigma(H)$ is the first common true level at or above H under $P_{x,y}^{++}$, then

$$\sigma(H) - H, \quad Y_{\sigma(H)}^{(i)} - Y_H^{(i)} \quad (i = 1, 2) \quad \text{are tight uniformly in } H, x, y. \quad (6)$$

At $\sigma(H)$ the pair suffix has the shared conditioned law from its new sites.

Proof. The probability that a deterministic level h is true under the single conditioned law is u_h , by the exact-truth identity in Section 2. Independence therefore makes the renewal mass for common words equal to $u_h^2 \geq p^2$. The renewal-count strong law, including its infinite-mean version obtained by truncation, implies $\bar{L} \leq p^{-2}$. The lower bound follows from $L \geq 1$.

For each of the two paths, let N_i be the number of its single words used before the first positive common boundary. Since single widths are positive integers, $N_i \leq L$. The event $N_i \geq j$ is determined by that path's first $j - 1$ words and the independent boundary sequence of the other path. It is consequently independent of the radius of its j -th word. The finite mean single-word radius from Section 2 gives

$$\mathbb{E} \sum_{j=1}^{N_i} R_{s,j}^{(i)} = \mathbb{E}N_i \mathbb{E}R_s \leq \bar{L} \mathbb{E}R_s < \infty.$$

The absolute value of S is bounded by the sum of these two total radii. Finally, uniform ellipticity permits two short paths with different second-coordinate displacements to level one. Imposing fresh truth there gives positive probability of an exact common cut with nonzero S .

We record the renewal estimate that supplies local tightness. For either the single renewal sequence or the independent common sequence, let L' be its word width and A any almost surely finite size of the word, such as its maximal displacement. If A_{word} is the size of the word based at the last boundary at or below a deterministic level h , then

$$\Pr(A_{\text{word}} > s) \leq \sum_{j=0}^h \Pr(L' > j, A > s) \leq \mathbb{E}[L' \mathbf{1}_{\{A > s\}}].$$

To obtain the first inequality, sum over the possible preceding boundary levels $h - j$, use independence of the next word, and bound each renewal mass by one. Since $\mathbb{E}L' < \infty$, the final quantity tends to zero as $s \rightarrow \infty$. The same reasoning with $A = L'$ gives tightness of the covering width. Thus first-hit positions at a deterministic layer differ from their preceding boundary positions by tight amounts. A fixed interval of integer levels meets only the words covering those levels, so it also has tight total displacement, uniformly in its base. Shared marginals inherit this statement from their uniform domination by the single conditioned law.

It remains to prove (6). Test the m levels

$$h = H, H + B, \dots, H + (m - 1)B.$$

Call failure at h *visible* if, after first hitting h , some walk drops below it before its first hit of $h + B$. In the raw law, the probability that all m tests have visible failure is at most $(1 - \underline{p}_2)^m$. Indeed, at each pair of first-hit arrivals to h , all previous visible-failure tests are already determined, while both fresh truths at h would preclude the next failure. Passing to the conditioned pair law costs at most $\underline{p}_2^{-1}$.

If h is not common true but has no visible failure, some walk drops below h after its first hit of $h + B$. That walk has no single true boundary between h and $h + B$, inclusive; its word covering h therefore has width greater than B . The covering-word estimate and marginal domination bound this alternative by

$$C\mathbb{E}[L_s \mathbf{1}_{\{L_s > B\}}].$$

A union bound over the m tests now gives

$$P_{x,y}^{++}(\sigma(H) > H + (m - 1)B) \leq \underline{p}_2^{-1}(1 - \underline{p}_2)^m + Cm\mathbb{E}[L_s \mathbf{1}_{\{L_s > B\}}].$$

Choose m large and then B large. This proves uniform tightness of the height overshoot. The fixed-interval displacement bound then proves the other assertions in (6). Finally, the first common boundary at or above H is a stopping index of the common-word chain, so its suffix rule follows from Lemma 3.1. $\square$

3.2 Truncated variance and median centering

For $r > 0$, define

$$m(r) = \mathbb{E}(S^2 \wedge r^2), \quad n(r) = r^2/m(r).$$

These quantities are positive by $\Pr(S \neq 0) > 0$. The scale $n(r)$ uses only the finite first moment of S ; a finite second moment is not assumed.

Lemma 3.3 (Scale and maximal estimates). *The function n is continuous and nondecreasing, tends to infinity, and is strictly increasing for all sufficiently large r . Write r_s for its inverse on large scales, so $n(r_s) = s$. Then*

$$n(cr) \geq c^2 n(r) \quad (0 < c \leq 1), \quad n(r) \geq r/\mathbb{E}|S|. \quad (7)$$

For iid copies S_j , let

$$M_H^S = \max_{0 \leq i \leq H} \left| \sum_{j=1}^i S_j \right|.$$

Uniformly in integer $H \geq 0$ and $z > 0$,

$$\mathbb{E}[(M_H^S)^2 \wedge z^2] \leq CHm(z), \quad \Pr(M_H^S > z) \leq CH/n(z). \quad (8)$$

Proof. The ratio $m(r)/r^2 = \mathbb{E}[(S^2/r^2) \wedge 1]$ is continuous and nonincreasing. There is a bounded interval of strictly positive values of $|S|$ having positive probability; its contribution is strictly decreasing once r exceeds that interval. This proves the monotonicity and eventual strict monotonicity of n . Monotonicity of m proves the first inequality in (7), and $S^2 \wedge r^2 \leq r|S|$ proves the second, hence $n(r) \rightarrow \infty$.

For (8), replace S_j by $S_j \mathbf{1}_{\{|S_j| \leq z\}}$. These truncated variables have mean zero by symmetry. The probability of any replacement among the first H variables is at most $H \Pr(|S| > z)$. Off this event the original partial sums agree with the truncated sums, whose expected squared maximum is at most $4H\mathbb{E}[S^2 \mathbf{1}_{\{|S| \leq z\}}]$ by the square maximal inequality. On the replacement event, the capped squared maximum is at most z^2 . Adding these bounds gives the first inequality in (8); Markov's inequality for the capped square gives the second. $\square$

Lemma 3.4 (Exact-level comparison). *For every integer $H \geq 0$ and $z > 0$,*

$$P_0^+(W_H > z) \leq C \Pr(M_H^S > z). \quad (9)$$

No moment of the maximal decoration inside a common word is needed.

Proof. In a raw first walk, select the first level $h \leq H$, if any, such that $T_h < T_{-1}$ and $|Y_h - b_h| > z$. Let A be the event that such a level exists. Conditional on this prefix and its selected level, an independent raw second walk reaches h before dropping, with its displacement on the opposite side of the median b_h , with probability at least $p/2$. Here the conditioned no-drop law only lower bounds a prefix probability; it is not imposed as a separate test on that second walk.

At the two first-hit endpoints, impose both fresh truths. Their probability factor is p^2 . The resulting event implies the two initial no-drop events and has h as an exact common true level, with $|\Delta_h| > z$. Dividing its raw probability by p^2 , its probability under the independent conditioned pair law is at least $P_0(A)p/2$. At a common cut of height at most H , the difference is a partial sum of at most H common-word increments, so

$$\Pr(M_H^S > z) \geq \frac{p}{2} P_0(A) \geq \frac{p^2}{2} P_0^+(W_H > z).$$

This proves (9). All of these adaptive prefix transfers can also be obtained by summing the corresponding finite stopped-word identities separately over the possible levels h . $\square$

Lemma 3.5 (Median linearization on short blocks). *For sufficiently small $t > 0$, uniformly in $1 \leq H \leq tn(r)$,*

$$\max_{j \leq H} |b_j - jb_H/H| \leq C\sqrt{tr}. \quad (10)$$

Proof. By (7)–(9), choose a fixed large C_0 , then t small enough that $C_0\sqrt{t} \leq 1$, to obtain

$$P_0^+(W_{2H} > C_0\sqrt{tr}) < p/4.$$

For $i + j \leq 2H$, the event that i is exactly true under P_0^+ has probability $u_i \geq p$, and its suffix has the fresh conditioned law. There is therefore positive probability on this event that Y_i , Y_{i+j} , and $Y_{i+j} - Y_i$ are all within $C_0\sqrt{tr}$ of b_i , b_{i+j} , and b_j , respectively. For the first two requirements use the preceding maximal bound, and for the third use the same bound on the fresh suffix. Consequently,

$$b_{i+j} = b_i + b_j + O(\sqrt{tr}) \quad (i + j \leq 2H),$$

uniformly in the indices.

Set $d_j = b_j - jb_H/H$ for $0 \leq j < H$. Apply the last relation to $j + j$, and, when $2j \geq H$, also to $H + (2j - H)$. This gives

$$2d_j = d_{2j \bmod H} + O(\sqrt{tr})$$

with a uniform error. Taking the maximum absolute value over $0 \leq j < H$ bounds that maximum by half itself plus $C\sqrt{tr}$. Since the error at $j = H$ is zero, this proves (10). $\square$

3.3 Change of conditioning and short-block survival

The estimates above concern P_x^+ , which conditions the joint annealed law on no drop. Under that law environments are weighted by $q(x)/p$. For quenched kernel estimates we also need the mixture

$$\tilde{P}_x = \int P_{x,\omega}(\cdot \mid D) P(d\omega),$$

which keeps the original environment law and conditions the path in each environment separately. We first transfer large-scale fluctuation bounds to this mixture. For blocks of height $H \leq tn(r)$, a fresh restart after the first loss of mass will retain positive kernel mass outside an environment exception of probability $O(t^2)$.

Lemma 3.6 (Transfer between conditioned laws). *Along any sequences of integers $H \geq 0$ and real numbers $z \rightarrow \infty$,*

$$\limsup \tilde{P}_0(W_H > z) \leq C \limsup P_0^+(W_H > z/C) \quad (11)$$

with a fixed constant C . If also $H \rightarrow \infty$, the same bound holds with the left-hand side evaluated under the P -average of the normalized successful-prefix kernel K_H/a_H at the origin.

Proof. On the joint environment-and-path space,

$$\frac{d\tilde{P}_0}{dP_0^+} = \frac{p}{q(0)}.$$

This density is integrable but need not be bounded. For any prescribed error, truncate it and then approximate the truncation by a bounded nonnegative environment function g depending on finitely many rows. Thus g approximates $p/q(0)$ in $L^1(P_0^+)$, and E_0^+g tends to one as the approximation error tends to zero. We will remove the finite initial segment that can be affected by g .

First check the deterministic centering error caused by such a removal. If $\limsup P_0^+(W_H > z/10) \geq 0.3$, Equation (11) is immediate with a sufficiently large fixed C . Otherwise these probabilities are eventually less than 0.3. For every fixed integer s_0 , local tightness then gives

$$|b_{h+s} - b_h| \leq 0.3z \quad (0 \leq s \leq s_0, h + s \leq H)$$

for all sufficiently large indices in the sequence. Indeed, each of $|Y_h - b_h| \leq z/10$ and $|Y_{h+s} - b_{h+s}| \leq z/10$ has probability greater than 0.7. By Lemma 3.2, the probability of $|Y_{h+s} - Y_h| \leq z/10$ tends to one uniformly over these indices. The three events have a nonempty intersection, which proves the claimed deterministic inequality by the triangle inequality.

Let s be the first positive single true boundary strictly above every row used by g . Its index in the single-word chain is a finite stopping index. For a sufficiently accurate approximation, $E_0^+g > 0$. Under the probability measure with density $g/(E_0^+g)$ relative to P_0^+ , the translated suffix at s has law P_0^+ , independently of the prefix. To verify this, apply exact-word factorization separately to each finite prefix through the selected boundary. Both its path weight and g use only rows below that boundary; all rows for the conditioned suffix are fresh. This remains true for rows used by g that the prefix has not visited. The separation is by height, not by which rows the walk happened to reveal.

For fixed g , first choose a deterministic s_0 so large that $s \leq s_0$ except for an arbitrarily small g -weighted error. As $z \rightarrow \infty$, we may further restrict to

$$\max_{j \leq s} |Y_j| \leq z/10, \quad \max_{j \leq s_0} |b_j| \leq z/10,$$

with a further error tending to zero. On these restrictions no offense $|Y_j - b_j| > z$ can occur before s . After s , write the relative suffix displacement as $Y'_u = Y_{s+u} - Y_s$. The centering estimate above gives, for $s < j \leq H$,

$$|Y_j - b_j| \leq |Y_s| + |Y'_{j-s} - b_{j-s}| + |b_{j-s} - b_j| \leq 0.4z + |Y'_{j-s} - b_{j-s}|.$$

Thus $W_H > z$ requires the fresh suffix to have its own $W_H > z/10$. If $H \leq s$, no offense remains at all.

The g -weighted probability is consequently bounded in the limsup by $E_0^+ g$ times $\limsup P_0^+(W_H > z/10)$, plus the restriction errors. Return to $\tilde{P}_0$ using the L^1 density approximation, and send all approximation and restriction errors to zero. Enlarging the same fixed C to include the earlier trivial case proves Equation (11).

Finally, in a fixed environment the event D is a subevent of $\{T_H < T_{-1}\}$. On first-hit prefixes, changing between these two conditional laws costs total variation at most $1 - q(0)/a_H$. Therefore the distance between $\tilde{P}_0$ on those prefixes and the P -average of K_H/a_H is at most

$$E(1 - q(0)/a_H) \rightarrow 0,$$

since $a_H \downarrow q(0) > 0$ almost surely. This proves the last assertion. $\square$

Proposition 3.7 (Short-block survival). *For every $v_0 > 0$ there are constants $C_{v_0} < \infty$ and $t_{v_0} > 0$ such that, for each fixed $0 < t \leq t_{v_0}$, some $\delta > 0$ satisfies, for all sufficiently large r , every integer $1 \leq H \leq tn(r)$, and every probability measure π on a starting layer,*

$$P\left(\pi K_H\{W_H \leq v_0 r, \max_{j \leq H} |Y_j - j b_H/H| \leq v_0 r\} < \delta\right) \leq C_{v_0} t^2. \quad (12)$$

The environment is fresh with product law, and path displacements are measured from each path's own starting site.

Proof. Fix $v_0 > 0$, set $w_0 = v_0/8$, and write $\theta = b_H/H$. The proof uses one estimate for loss of mass in a narrower slope tube, then restarts that estimate after its first failure. All stopping tests will use finite-height kernels.

One possible loss of mass. Choose t_{v_0} small enough that Equation (10) gives, for sufficiently large r , uniformly over the stated H ,

$$\max_{j \leq H} |b_j - j\theta| \leq w_0 r/2, \quad |\theta| \leq w_0 r.$$

The second inequality follows from the first at $j = 1$, since b_1 is fixed. Equations (7)–(9) and (11) also give

$$E[P_{0,\omega}(W_H > w_0 r/2 \mid D)] \leq C_{v_0} t$$

uniformly in $1 \leq H \leq tn(r)$ for large r . Indeed, a violating sequence of H, r would contradict the limsup transfer in Equation (11), whose right side is bounded by a constant times $H/n(r)$. The scale inequality in Equation (7) absorbs the fixed change in threshold.

Choose $\delta_0 \in (0, 1/4)$ so that $P(q(0) < 4\delta_0) \leq t$. Call a starting site x favorable when

$$q(x) \geq 4\delta_0, \quad P_{x,\omega}(W_H \leq w_0 r/2 \mid D) \geq 1/2.$$

The expected π -fraction of unfavorable sites is at most $C_{v_0} t$. Markov's inequality shows that, except on an environment event of probability at most $C_{v_0} t$, favorable sites carry at least half the initial mass. From each favorable site the raw mass of paths on D in the median

tube is at least $2\delta_0$. Their first-hit prefixes lie in the slope tube of width w_0r at every shorter horizon. Hence, except on that one environment event,

$$M_u^\pi := \pi K_u \left\{ \max_{j \leq u} |Y_j - j\theta| \leq w_0r \right\} \geq \delta_0 \quad (0 \leq u \leq H).$$

In particular the probability that any of these inequalities fails is at most $C_{v_0}t$, uniformly in the initial probability π . No independence among the environments seen from different starting sites was used.

A fresh restart after the first failure. Test the masses M_u^π in increasing order, and let s be the first level at which the displayed inequality fails. There is no failure at level zero, since $M_0^\pi = 1$. Each test is measurable from rows strictly below its tested layer, so $s \leq H$, when finite, is an environment stopping layer.

The passing submeasure at layer $s - 1$ has mass at least δ_0 . Extend every one of its prefixes by one direct $+e_1$ step. Uniform ellipticity leaves mass at least $\kappa\delta_0$. The extension has slope error at most $2w_0r$, since the earlier error is at most w_0r and $|\theta| \leq w_0r$. Both detection of the failure and this bridge use only rows below layer s .

Normalize the bridged endpoint submeasure to a probability π_s and restart at layer s . Conditional on the exposed lower rows, π_s is fixed and the upper field has its original product law. The one-failure estimate therefore applies again. Use the same slope $\theta = b_H/H$ and its original horizon H ; its simultaneous conclusions imply the required ones through the shorter remaining horizon $H - s$. Replacing H by $H - s$ in the slope would not be justified and is unnecessary.

Thus, conditional on the first failure, the probability of a second failure before the remaining horizon is at most $C_{v_0}t$. The probability of two failures is at most $C_{v_0}t^2$, after enlarging the constant. This multiplication uses fresh upper rows at the stopping layer, not independence of two events in the original slab.

Retained mass and its tube. If no first failure occurs, at least δ_0 of the original kernel mass reaches the top in the slope tube. If there is a first failure but no second one, concatenate the bridged mass and the restarted successful prefixes. Retain each prefix's original starting site when doing so; restarted displacements are measured from the new endpoint. The total slope error is at most $2w_0r + w_0r = 3w_0r$, and the retained mass is at least $\kappa\delta_0^2$. Local no-drop successes are also successes above the original floor, so the concatenated kernels are submeasures of the original K_H .

Adding the deterministic median-line error $w_0r/2$ puts these prefixes in the required median tube as well. Both tube widths are less than v_0r . Take $\delta = \min\{\delta_0, \kappa\delta_0^2\}$ to obtain Equation (12). The infinite no-drop event served only to bound failure probabilities; every test, bridge, and restart used the finite-height prefix kernels. $\square$

3.4 Endpoint spread

Lemma 3.8 (Spread at the truncated-variance scale). *Under the independent conditioned pair law, there is a constant $c > 0$ such that*

$$\begin{aligned} \Pr(|\Delta_{\lfloor n(r) \rfloor}| > cr) &\geq c \quad (r \text{ large}), \\ \liminf \Pr(|\Delta_{\lfloor tn(r) \rfloor}| > lr/4) &\geq c_{l,\epsilon}t \quad \text{if } r \rightarrow \infty, \quad n(r) \Pr(|S| > lr) \geq \epsilon. \end{aligned} \tag{13}$$

In the second assertion $0 < l < 1$ and $\epsilon > 0$ are fixed, and the bound holds for each sufficiently small fixed $t > 0$, depending on l, ϵ . Relative to any deterministic endpoint center, a single P^+ -walk consequently has a comparable probability of a deviation, at half the corresponding separation threshold, in at least one of the two signs.

Proof. Fix $s > 0$, put $H = \lfloor sn(r) \rfloor$, and let $k = \lfloor H/\bar{L} \rfloor$. First we justify the approximation

$$\Delta_H - \sum_{i=1}^k S_i = o_{\text{Pr}}(r).$$

By Lemma 3.2, the displacement from the preceding common boundary to the first-hit position at H is tight. The renewal strong law puts the number of completed common words within $o_{\text{Pr}}(n(r))$ of k . No independence between this count and the increments is required: on the event that its distance from k is at most $an(r)$, bound the sum error by the maximal sums in the deterministic forward and backward windows of that length around k . For every fixed $\rho > 0$, (7)–(8) bound the probability of such an error exceeding ρr by $O_\rho(a) + o(1)$. Let $a \downarrow 0$, and include the tight decoration error, to obtain the approximation.

For $s = 1$, set

$$V = \sum_{i=1}^k (S_i^2 \wedge r^2).$$

Its mean is comparable to r^2 , since $k \sim n(r)/\bar{L}$, and

$$\mathbb{E}V^2 \leq kr^2m(r) + k^2m(r)^2 = O(r^4).$$

The second-moment inequality therefore makes V exceed a fixed positive multiple of r^2 with uniformly positive probability. Conditional on the magnitudes $(|S_i|)$, the nonzero signs are independent fair signs. The sum $\sum_{i=1}^k S_i$ has conditional variance $Q = \sum_{i=1}^k S_i^2 \geq V$ and conditional fourth moment at most $3Q^2$. A further second-moment inequality gives a uniformly positive conditional probability of a displacement of order $\sqrt{Q}$, hence of order r on the preceding event. The approximation, with a smaller displacement constant, proves the first line of (13).

For the second line, take $s = t$ and set $p_r = \Pr(|S| > lr)$. The assumptions and the definition of m give

$$\epsilon \leq n(r)p_r \leq l^{-2}.$$

Among $k \sim tn(r)/\bar{L}$ summands, the probability that exactly one has magnitude greater than lr is therefore at least $c_{l,\epsilon}t$ for each sufficiently small fixed $t > 0$ and large r . Conditional on the index of that exceptional summand, the others are independent symmetric variables conditioned to have magnitude at most lr . Their sum has variance at most

$$\frac{k \mathbb{E}[S^2 \mathbf{1}_{\{|S| \leq lr\}}]}{1 - p_r} \leq Ctr^2.$$

For t sufficiently small, Chebyshev's inequality gives probability at least $1/2$ that their sum has magnitude at most $lr/2$. On this event they cannot cancel more than half the exceptional summand. Thus the deterministic sum has magnitude greater than $lr/2$ with probability at least a constant times t . Keep t fixed while $r \rightarrow \infty$; the $o_{\text{Pr}}(r)$ approximation then proves the claimed bound at threshold $lr/4$.

Finally, if two iid endpoint displacements differ by more than a , at least one differs from any given deterministic center by more than $a/2$. The probability of this single-walk absolute deviation is at least half the two-walk separation probability. One of its two signs has at least half that mass. This proves the final assertion after adjusting constants. $\square$

4 Gaussian scales

We next identify the fluctuation law on scales where the independent common-word increments have negligible large jumps. There are three steps: an annealed limit for independent paths,

a comparison for shared paths away from one another, and an occupation estimate excluding a singular interaction on their projected diagonal. The resulting shared two-copy limit will imply a quenched limit in probability.

Definition 4.1 (Gaussian sequence). A sequence $r \rightarrow \infty$ is *Gaussian* if

$$n(r) \Pr(|S| > ar) \rightarrow 0 \quad \text{for every fixed } a > 0. \quad (14)$$

Throughout this section limits are along any fixed Gaussian sequence. Height-indexed processes are linearly interpolated on the grid $h/n(r)$. Functional convergence means weak convergence in $C([0, T], \mathbb{R}^k)$, with the uniform norm, for every fixed finite T ; a slightly larger horizon is used when interpolating near its endpoint. Put $\sigma_*^2 = 1/(2\bar{L})$.

4.1 Independent-path Brownian limit

The common-word increments give a Brownian limit at common cuts. To obtain a limit for a single walk at every first-hit level, we must also control the fluctuations between cuts and recover the deterministic centering. Neither step requires a second moment for word decorations.

Lemma 4.2 (Independent first-hit fluctuations). *For two independent conditioned paths, Δ_h/r converges functionally to Brownian motion of variance parameter $1/\bar{L}$. For a single P_0^+ path, $(Y_h - b_h)/r$ converges functionally to Brownian motion of variance parameter σ_*^2 .*

Proof. Write (L_i, S_i) for the successive independent common-word widths and difference increments, and put

$$C_m = \sum_{i=1}^m L_i, \quad A_m = \sum_{i=1}^m S_i, \quad N(h) = \max\{m : C_m \leq h\}.$$

The difference at the preceding common cut is $\hat{\Delta}_h = A_{N(h)}$. As with the first-hit process, we interpolate its values linearly on the height grid $h/n(r)$.

The limit at common cuts. First consider A_m/r , with $n(r)$ summands per unit time. The Gaussian-sequence assumption permits a symmetric truncation

$$S_i^{(r)} = S_i \mathbf{1}_{\{|S_i| \leq \eta_r r\}}, \quad \eta_r \downarrow 0,$$

for which

$$n(r) \Pr(|S| > \eta_r r) \rightarrow 0, \quad \frac{\mathbb{E}[(S_i^{(r)})^2]}{m(r)} \rightarrow 1.$$

Indeed, for every fixed $0 < \eta < 1$,

$$0 \leq m(r) - \mathbb{E}[S^2 \mathbf{1}_{\{|S| \leq \eta r\}}] \leq r^2 \Pr(|S| > \eta r) = o(m(r));$$

a sufficiently slow diagonal choice of η_r gives both requirements. The probability that any summand is removed on a fixed time interval tends to zero.

The variables $S_i^{(r)}/r$ have mean zero, variance $(1 + o(1))/n(r)$, and fourth moment at most η_r^2 times that variance. Taylor expansion of their characteristic functions therefore gives the standard Brownian finite-dimensional laws. For tightness, the fourth-moment computation for independent centered sums, followed by the fourth-moment maximal inequality, bounds the expected maximal fourth power on a block of $\lfloor \delta n(r) \rfloor$ summands by

$$O(\delta^2 + \delta \eta_r^2).$$

A union bound over the blocks in a fixed time interval, followed by $\delta \downarrow 0$, proves tightness. Thus A_m/r has the standard Brownian functional limit.

The width strong law gives

$$\sup_{0 \leq h \leq Tn(r)} \left| \frac{N(h)}{n(r)} - \frac{h}{\bar{L}n(r)} \right| \rightarrow 0 \quad \text{in probability.}$$

Applying this time change to the tight sum process, whose limits are continuous, proves the Brownian limit for $\hat{\Delta}_h/r$, with variance parameter $1/\bar{L}$. The counts and the sums need not be independent. The maximal normalized jump tends to zero as well, so the stated interpolation convention has the same limit.

Passing from cuts to every first hit. Lemma 3.2 makes $\Delta_h - \hat{\Delta}_h$ tight uniformly at deterministic levels h . This already gives the required finite-dimensional limits. To prove path tightness, fix $\delta > 0$ and partition the height range into consecutive blocks of $\lfloor \delta n(r) \rfloor$ levels. There are only finitely many block starts for fixed T, δ .

Suppose that within one of these blocks, with start a ,

$$|\Delta_h - \Delta_a| > \rho r$$

for some level h in the block. Under the raw independent pair law, select the first such offending level among prefixes for which both walks have avoided the initial drop. The selection is determined by the paired prefixes through that level. The raw probability of finding an offense is at least p^2 times its probability under the independent conditioned pair law. Imposing both fresh truths at the selected level multiplies each selected prefix weight by p^2 , leaves the offense unchanged, and implies both initial no-drop events. After division by the initial normalizer p^2 , both probabilities below are under the conditioned pair law, and

$$\Pr(\text{an offense}) \leq p^{-2} \Pr(\text{an offense at a common cut}).$$

The same statement holds for the union of the blocks by selecting its first offense.

At a common cut h , the offending difference satisfies

$$|\Delta_h - \Delta_a| \leq |\hat{\Delta}_h - \hat{\Delta}_a| + |\Delta_a - \hat{\Delta}_a|.$$

The maximum of the second term over the finitely many deterministic block starts is $o_{\Pr}(r)$. The first term is controlled by the modulus of the common-cut process, which has a continuous Brownian limit. Consequently, for each $\rho > 0$, the limsup probability of an offense tends to zero as $\delta \downarrow 0$. Adjacent-block bounds and interpolation give tightness of Δ_h/r , proving the first assertion.

Recovering a single path. Let μ_r be the law in continuous path space of the interpolated single-path process Y_h/r , on a fixed compact time interval. Its iid differences are tight by the first assertion. We first show that μ_r is tight after suitable deterministic path translations. Choose compact sets K_j capturing the difference laws with errors ε_j^2 , where $\varepsilon_j \downarrow 0$ and $\sum_j \varepsilon_j < 1$. If a prospective center f is sampled from μ_r , then

$$\int \mu_r(f + K_j) \mu_r(df) \geq 1 - \varepsilon_j^2.$$

Markov's inequality shows that the centers for which $\mu_r(f + K_j) < 1 - \varepsilon_j$ have μ_r -probability at most ε_j . Some center path f_r therefore works for every j , and the translated laws are tight. We may choose f_r with the same interpolation as the original paths.

Along a subsequence on which these translated laws converge, let G be the continuous limiting process. Its iid two-copy difference has the Brownian law just proved. Cramér's

normal decomposition principle [6], in the iid-difference form verified below, applied to every scalar linear combination of finitely many coordinates of G , shows that G has Gaussian finite-dimensional laws. Its covariance is half that of the difference process, namely

$$\text{Cov}(G(s), G(t)) = \sigma_*^2 \min(s, t).$$

Only a deterministic mean function remains to be identified.

For completeness, suppose that $Z - Z'$ is normal for iid real variables Z, Z' . Choose C_0 with $\Pr(|Z'| \leq C_0) > 0$. Intersecting either tail of Z with this event bounds that tail by a constant times a Gaussian tail. Hence $f(z) = \mathbb{E}e^{zZ}$ is entire and obeys $|f(z)| \leq \exp(C(1 + |z|^2))$. The identity $f(z)f(-z) = \mathbb{E}e^{z(Z-Z')}$, first on the real axis and then everywhere, shows that f has no zeros. Its entire logarithm g , chosen with $g(0) = 0$, satisfies $\text{Re } g(z) \leq C(1 + |z|^2)$. On every circle the mean of $\text{Re } g$ is zero, so its absolute integral mean is also $O(1 + |z|^2)$. The Fourier coefficient formula on circles of arbitrarily large radius then makes every Taylor coefficient of g of order greater than two vanish. Thus g is quadratic, real on the real axis, with $g''(0) = \text{Var } Z \geq 0$. It follows that Z is normal, allowing a point mass.

Write $m_G(t) = \mathbb{E}G(t)$. Sample continuity and convergence in law of Gaussian marginals at convergent times imply that m_G is continuous. Each marginal has the unique median $m_G(t)$, including at time zero. At a grid time $h/n(r)$, the number

$$\frac{b_h}{r} - f_r(h/n(r))$$

is a median of the translated marginal. Along every sequence of grid times converging to t , weak path convergence and uniqueness of the limiting median make these medians converge to $m_G(t)$. Compactness of the time interval makes the convergence uniform over grid times; interpolation gives uniform convergence on the whole interval. Use a slightly longer interval when interpolating at the endpoint. Subtracting the interpolated medians from the original process therefore removes precisely the limiting mean function. Every subsequential limit is Brownian motion of variance parameter σ_*^2 , proving the single-path assertion. $\square$

Lemma 4.3 (Linearization of the medians). *Uniformly for $k + j \leq Tn(r)$, for each fixed T ,*

$$b_{k+j} - b_k - b_j = o(r). \quad (15)$$

Writing $N_r = \lfloor n(r) \rfloor$ and $\theta_r = b_{N_r}/N_r$, we have

$$\max_{h \leq Tn(r)} |b_h - h\theta_r| = o(r), \quad \theta_r = o(r).$$

Proof. We first prove the additive relation. It suffices to consider arbitrary sequences of indices for which $k/n(r) \rightarrow s$ and $j/n(r) \rightarrow t$, and to pass further so that k is fixed or tends to infinity. Let $\mathcal{C}_k$ be exact single truth at level k . Under P_0^+ , this event has probability $u_k \geq p$. Conditional on $\mathcal{C}_k$, the prefix has the raw law conditioned on $T_k < T_{-1}$, and its relative suffix is an independent fresh P_0^+ path.

If $k \rightarrow \infty$, the total variation distance between this prefix law and the P_0^+ prefix law is at most $1 - p/u_k \rightarrow 0$. If k is fixed, its displacement and median are fixed random and deterministic quantities, respectively, and their normalized difference tends to zero. Lemma 4.2 therefore gives, under the conditional law,

$$\frac{Y_{k+j} - b_k - b_j}{r} \implies N(0, \sigma_*^2(s + t)).$$

The same lemma gives this normal limit under the unconditional P_0^+ law when the centering is b_{k+j} .

Set $d_r = (b_k + b_j - b_{k+j})/r$. The unconditional distribution centered by b_{k+j} dominates p times the conditional distribution with that same centering. Tightness first forces d_r to be bounded. Along a further subsequence with $d_r \rightarrow d$, the limiting domination would read

$$N(0, v) \geq pN(d, v), \quad v = \sigma_*^2(s + t),$$

as measures. For $v > 0$, a nonzero shift makes the ratio of the shifted normal density to the unshifted one unbounded. For $v = 0$, the two point masses have different supports unless $d = 0$. Thus $d = 0$ in every case. Compactness of the ranges of $k/n(r)$ and $j/n(r)$ proves the uniform relation (15).

To turn approximate additivity into a linear approximation, put $N = N_r$, $\theta = b_N/N$, and $d_j = b_j - j\theta$ for $0 \leq j < N$. Apply (15) to $j + j$, and also to $N + (2j - N)$ when $2j \geq N$. Uniformly over these indices,

$$2d_j = d_{2j \bmod N} + o(r).$$

The maximum of $|d_j|$ is therefore at most half itself plus $o(r)$, so it is $o(r)$. For $h \leq Tn(r)$, write $h = qN + j$, where q remains bounded, and apply (15) a bounded number of times. This proves $\max_{h \leq Tn(r)} |b_h - h\theta_r| = o(r)$. Finally, at $h = 1$ this gives $|b_1 - \theta_r| = o(r)$; since b_1 is fixed, $\theta_r = o(r)$. $\square$

4.2 Shared paths: marginals and separation

Two paths in one environment interact when their traces visit a common site. We first show that the conditioning on the second path does not change the limiting fluctuations of the first. We then bound trace contact by the probabilities of missing two disjoint endpoint windows. This comparison concerns the entire traces, including their excursions between first hits of successive layers.

Lemma 4.4 (Shared marginal limits). *Under $P_{x,y}^{++}$, each marginal process*

$$B_i^r(h/n(r)) = (Y_h^{(i)} - h\theta_r)/r \tag{16}$$

converges functionally to Brownian motion of variance parameter σ_^2 , uniformly over starts on a common layer. Thus the conclusion holds along every choice $x = x(r), y = y(r)$ with $x_1 = y_1$, whatever their separation. The pair is jointly tight.*

Proof. Relative to the single conditioned environment-and-path law from x , the first marginal has density

$$\frac{p q(y)}{p_2(x, y)}.$$

For a fixed integer J , replace $q(y)$ by $a_J(y)$, then normalize. The resulting probability law differs from the original one in total variation by at most $C(u_J - p)$, uniformly in x, y . Indeed $a_J \geq q$, and the added mass before normalization is

$$\frac{E[q(x)(a_J(y) - q(y))]}{p_2(x, y)} \leq \frac{u_J - p}{p_2}.$$

All these normalized laws have density bounded by a common constant relative to the single conditioned law.

The weight $a_J(y)$ uses only rows between the common base and the layer J above it, excluding the latter. Stop the single-word chain of the first walk at its first true boundary at or above that layer. Every row used by the weight lies below this boundary. Exact-word factorization therefore gives a fresh P^+ suffix, independent of the removed path prefix under

the weighted law. The weight may depend on infinitely many sites in those J layers; only their heights matter for this factorization.

For fixed J , the height and size of this initial prefix are tight under P_x^+ , and hence under every one of the bounded weighted laws. Its height divided by $n(r)$ tends to zero. Its displacement divided by r also tends to zero, as does its centering correction, since $\theta_r/r \rightarrow 0$. The suffix limit and its tight continuous modulus thus give the asserted limit under the approximating law. Send $r \rightarrow \infty$ first and then $J \rightarrow \infty$. The same proof applies to the second marginal, and tightness of the two marginals gives joint tightness. $\square$

The next elementary observation converts endpoint concentration into a bound on trace contact. Fix an environment and a base layer of height F . Let each walk be conditioned never to fall below F , and stop it on first arrival at height $F + k$. For disjoint sets I_1, I_2 in that top layer, write E_i for the event that walk i 's endpoint lies outside I_i . Then

$$P_{x,\omega}(\cdot \mid D) \otimes P_{y,\omega}(\cdot \mid D) \{\text{the two stopped traces meet}\} \leq 2P_{x,\omega}(E_1 \mid D) + 2P_{y,\omega}(E_2 \mid D).$$

Here the two occurrences of D use the same absolute floor F . To prove the bound, for $F \leq z_1 \leq F + k$ let $v_i(z)$ be the probability of a top endpoint in I_i for a quenched walk from z conditioned to avoid that floor. We consider only sites with positive probability of such avoidance. The Markov property makes $v_i(z)$ the conditional endpoint probability whenever either walk first visits z , regardless of which walk visits it. Since I_1, I_2 are disjoint, $v_1(z) + v_2(z) \leq 1$.

Put $A_i = \{z : v_i(z) \leq 1/2\}$. At the first visit to A_i before or at the top, the conditional probability of E_i is at least $1/2$. The quenched Markov property at that first visit gives

$$P_{x,\omega}(\text{visit } A_1 \text{ by the top} \mid D) \leq 2P_{x,\omega}(E_1 \mid D),$$

and the analogous inequality holds for the walk from y . Every common site belongs to at least one A_i . A union bound proves the contact bound; no synchronization of the two visit times is required.

Lemma 4.5 (Off-diagonal comparison). *Fix $a, t > 0$. Uniformly over same-layer starts with $|y_2 - x_2| \geq ar$, the total variation distance between the pair path-prefix laws through level $k = \lfloor tn(r) \rfloor$, under $P_{x,y}^{++}$ and under independent conditioned walks, satisfies*

$$\limsup \text{dist}_{\text{TV}} \leq C \exp(-ca^2/t). \quad (17)$$

The constants depend only on the walk law.

Proof. In the top layer choose the windows

$$I_1 = \{z : |z_2 - x_2 - k\theta_r| < ar/3\}, \quad I_2 = \{z : |z_2 - y_2 - k\theta_r| < ar/3\}.$$

They are disjoint. Lemma 4.4 and the normal tail bound show that the shared conditioned probability of either endpoint error has limsup at most $Ce^{-ca^2/t}$, uniformly in the starts. Average the contact bound under the environment law with density $q(x)q(y)/p_2(x, y)$. This gives the same bound for trace contact under $P_{x,y}^{++}$.

To compare laws off contact, first use raw successful prefixes: both walks reach level k before dropping below the common base. For any two disjoint finite traces, their raw prefix weights agree after averaging in shared and independent environments, because their departure-row sets are disjoint. Passing from two finite successes to two infinite no-drop events removes raw mass at most $2(u_k - p)$ under either pair law. Moreover $p_2(x, y) - p^2 \rightarrow 0$ uniformly for the present starts, since their separation tends to infinity. Thus the conditioned prefix laws agree off contact up to a variation error that tends to zero. Their total masses are both one, so the contact mass on the independent side has the same upper bound up to this error. The contact estimate proves (17). $\square$

4.3 Zero occupation on the projected diagonal

The preceding comparison applies when the second-coordinate gap is of order r . It does not exclude a limiting pair that spends positive time at zero gap. We now bound that occupation. The estimate counts common true cuts first; exact-truth factorization will then transfer it to every deterministic layer.

Proposition 4.6 (Vanishing diagonal occupation). *Along every Gaussian sequence, under $P_{0,0}^{++}$, for each fixed finite T ,*

$$\lim_{\rho \downarrow 0} \limsup_{r \rightarrow \infty} \frac{1}{n(r)} E_{0,0}^{++} \sum_{0 \leq h \leq Tn(r)} \mathbf{1}_{\{|Z_h| \leq \rho r\}} = 0. \quad (18)$$

Proof. Choose $1 < \alpha < \gamma < 2$. The potential will be a flattened and capped version of u^α , where u is the absolute gap at an observed common cut. Its expected increase will pay for the number of near-diagonal cuts before the next observation. We need one estimate at nearly Gaussian radii and another below the largest remaining exceptional radius.

Drift at nearly Gaussian radii. Fix a sufficiently small $t_0 > 0$. There exist $l \in (0, 1)$, $\epsilon > 0$, and $u_0 < \infty$ such that every radius satisfying

$$u \geq u_0, \quad n(u) \Pr(|S| > lu) \leq \epsilon$$

has, uniformly over shared starts with $|Z_0| = u$,

$$\begin{aligned} E^{++}[(|Z_{\sigma(k(u))}|/u)^\alpha \wedge 2^\alpha] &\geq 1 + c_* t_0, \\ P^{++}\left(\min_{0 \leq h \leq k(u)} |Z_h| \leq u/2\right) &\leq C_* e^{-c_*/t_0}, \quad k(u) = \lfloor t_0 n(u) \rfloor \geq 1. \end{aligned} \quad (19)$$

Call these radii good. The constants c_*, C_* can be fixed independently of small t_0 ; l, ϵ, u_0 may depend on it.

Here is the compactness argument giving this uniform assertion. Along any Gaussian sequence of radii u , the independent gap at height $k(u)$, divided by u , converges to $\pm 1 + G$, where $G \sim N(0, 2\sigma_*^2 t_0)$. For small t_0 ,

$$E[|1 + G|^\alpha \wedge 2^\alpha] \geq 1 + ct_0.$$

Indeed $|z|^\alpha$ has strictly positive second derivative near one, while the Gaussian probability of leaving that neighborhood is exponentially small in $1/t_0$. The off-diagonal comparison changes the expectation by at most Ce^{-c'/t_0} in the limit. Replacing $k(u)$ by $\sigma(k(u))$ costs $o(1)$: by (6) the unscaled gap difference is uniformly tight, and the tested function is bounded and uniformly continuous. No moment of the overshoot is used.

For the second estimate, shrinking the gap by $u/2$ requires at least one centered marginal to move by $u/4$. The shared marginal limit and the Brownian maximal bound

$$\Pr\left(\sup_{s \leq t_0} |B(s)| \geq x\right) \leq 2 \exp\left(-\frac{x^2}{2\sigma_*^2 t_0}\right)$$

give the required limit bound. This maximal bound follows by applying the nonnegative submartingale inequality to the two exponentials of Brownian motion on finite grids and then using continuity.

Choose strict slack in both limiting inequalities. If no triple (l, ϵ, u_0) gave (19), choose violating radii while $l, \epsilon \downarrow 0$ and $u_0 \rightarrow \infty$. These radii would form a Gaussian sequence, contradicting the preceding limits. Finally, decrease ϵ until $4/(1 + 4\epsilon) \geq 2^\gamma$, and increase u_0 until $k(u) \geq 1$.

A cutoff above all exceptional radii. Fix $A > 1$. Along the given Gaussian sequence r , the nongood radii in $[u_0, Ar]$ are all $o(r)$. For every fixed $\xi > 0$, this follows uniformly on $[\xi r, Ar]$ from the Gaussian tail tests, monotonicity of n , and $n(Ar) \leq A^2 n(r)$.

On any subsequence pass further as follows. If the nongood radii stay bounded, choose a fixed cutoff b_o above them and above u_0 . Otherwise choose a nongood radius $D_* \rightarrow \infty$ within a factor two of their supremum, and set $b_o = 4D_*$. In either case $b_o = o(r)$, every radius between b_o and Ar is good, and

$$\frac{n(u)}{n(r)} \leq C_A (u/r)^\gamma \quad (b_o \leq u \leq Ar). \quad (20)$$

For this last bound, at a good radius

$$m(2u) \leq m(u) + 4u^2 \Pr(|S| > u) \leq (1 + 4\epsilon)m(u),$$

so $n(2u) \geq 2^\gamma n(u)$. Iterate while the doubling stays below r . For $r \leq u \leq Ar$, monotonicity of m gives the remaining estimate. The constant C_A need not depend on t_0 .

Escape from the cutoff region. Uniformly over $|Z_0| \leq b_o$, there is a positive probability of reaching an exact common true level $h \leq K$ with gap at least $2b_o$, where $K \leq Cn(b_o)$. The constants are uniform along the chosen further sequence; when b_o is fixed they may depend on that fixed value.

For a fixed cutoff, bounded uniformly elliptic scripts to level one can enlarge the gap beyond $2b_o$; fresh joint truth completes the claim. Consider therefore $b_o = 4D_*$ with $D_* \rightarrow \infty$. Since D_* is nongood, the second line of (13), at $H = \lfloor t_1 n(D_*) \rfloor$, supplies one deterministic sign of single-path deviation from b_H , of size at least $c_0 D_*$ and P^+ -probability at least $c_1 t_1$. Here $t_1 > 0$ is fixed and sufficiently small, while c_0, c_1 do not depend on this small choice. The raw quenched mass of that deviation therefore exceeds $pc_1 t_1 / 2$ on an environment event of probability at least $pc_1 t_1 / 2$.

Assign this deviation to the rightmost walk for a positive sign, or to the leftmost walk for a negative sign. For the other walk, (12) supplies fixed positive raw mass within $c_0 D_*/3$ of b_H , except on an environment event of probability $O_{c_0}(t_1^2)$. Subtracting that error from the preceding $\Omega(t_1)$ probability gives a uniformly positive probability that both masses are available in the same environment. Multiplying the quenched masses then gives a uniformly positive raw probability of two successes whose absolute gap has increased by at least $c_0 D_*/2$. Their prefixes use only rows below the top. Fresh joint truth there has probability at least p_2 , so the top is an exact common cut and both initial no-drop events hold.

Repeat a fixed number of these blocks, choosing the outward-moving walk at each new common cut. The shared continuation rule applies at every repetition. Enough repetitions enlarge any initial gap in $[0, 4D_*]$ beyond $8D_*$, with uniformly positive probability, and their total height is a fixed multiple of H . This proves the claim with $K \leq Cn(b_o)$.

Observation times and potential increase. Starting at the common cut zero, observe only the following subset of common cuts. Stop when an observed gap is at least Ar . From an observed gap $u < Ar$, choose the next observation by the rule

$$\begin{cases} \text{first common cut at offset at least } k(u), & u > b_o, \\ \text{first subsequent common cut with gap at least } 2b_o \text{ or offset at least } K, & u \leq b_o. \end{cases}$$

These are stopping indices of the common-word chain. At each observation the conditional continuation is the fresh shared conditioned pair law from the observed sites.

Use the bounded potential

$$\Phi(u) = [(u \vee b_o) \wedge 2Ar]^\alpha.$$

For $b_o < u < Ar$, (19) implies an expected increase at least $c_* t_0 u^\alpha$. Flattening below b_o only increases the terminal value, and the upper cap is at least $2u$. For $u \leq b_o$, the potential cannot decrease; the escape event just proved makes it increase by at least $(2^\alpha - 1)b_o^\alpha$ with a uniformly positive probability. Telescope up to observed exit, first with a deterministic truncation of the number of observations. Since $\Phi \leq (2Ar)^\alpha$, monotone convergence gives

$$E^{++} \sum_{\substack{\text{pre-exit observations} \\ u > b_o}} t_0 u^\alpha \leq C(Ar)^\alpha, \quad E^{++} \sum_{\substack{\text{pre-exit observations} \\ u \leq b_o}} b_o^\alpha \leq C_o(Ar)^\alpha.$$

The first constant is independent of small t_0 .

Counting all common cuts. Allocate each common cut to the last observation at or before it, excluding the next observation. An upper-regime observation at gap u receives at most $k(u)$ cuts: each allocated offset is an integer strictly below $k(u)$. A lower-regime observation receives at most K cuts. These bounds do not require any bound on the height overshoot of the next observation.

Let N_ρ^{pre} count allocated common cuts with gap at most ρr , before observed exit. Split observations into three sets. For fixed $\rho > 0$, take r large enough that $b_o \leq 2\rho r$. Equation (20) and the drift bounds give

gap at the observation	contribution to $E^{++} N_\rho^{\text{pre}} / n(r)$
$u \leq b_o$	$C_{A,o}(b_o/r)^{\gamma-\alpha}$
$b_o < u \leq 2\rho r$	$C_A \rho^{\gamma-\alpha}$
$2\rho r < u < Ar$	$C_A e^{-c_*/t_0}$

For the first row use $K \leq Cn(b_o)$. For the second use $k(u) \leq t_0 n(u)$ and $u^{\gamma-\alpha} \leq (2\rho r)^{\gamma-\alpha}$. For the last row, an allocated near-diagonal cut requires shrinking the gap from u to at most $u/2$ before offset $k(u)$. Its conditional probability is bounded by (19); the number of allocated cuts is at most $k(u)$. In both upper-regime rows, the factor t_0 from $k(u)$ cancels the one in the drift budget. Their constants can therefore be independent of small t_0 .

Under the same-site initial law, joint marginal tightness makes the probability of an observed gap $\geq Ar$ before height $Tn(r)$ tend to zero as $A \rightarrow \infty$, uniformly in the limiting upper bound in r . All common cuts after that exit and below the target height contribute at most $Tn(r) + 1$ times its indicator.

From common cuts to deterministic layers. Let $\mathcal{C}_h$ denote common truth at height h , and let A_h be the raw probability that both paths reach h before the initial drop with $|Z_h| \leq \rho r$. Imposing the initial shared conditioning gives a numerator at most A_h . Instead, impose both fresh truths at height h . Their conditional probability is at least $\underline{p}_2$, and the resulting event implies the initial conditioning and $\mathcal{C}_h$. Dividing by the same initial normalizer gives

$$P_{0,0}^{++}(|Z_h| \leq \rho r) \leq \underline{p}_2^{-1} P_{0,0}^{++}(|Z_h| \leq \rho r, \mathcal{C}_h).$$

Sum over $h \leq Tn(r)$ and use the common-cut bounds above. On the chosen further sequence the lower-regime term vanishes because $b_o = o(r)$. The remaining pre-exit bound is $C_A \rho^{\gamma-\alpha} + C_A e^{-c_*/t_0}$. Every subsequence admits this further-sequence construction, so the bound controls the original limsup. Send $\rho \downarrow 0$, then $t_0 \downarrow 0$, and finally $A \rightarrow \infty$. This proves (18). $\square$

4.4 Shared joint limit and quenched consequence

Proposition 4.7 (Independent joint limit for shared paths). *Under $P_{0,0}^{++}$, the pair of processes in (16) converges functionally to two independent Brownian motions, each with variance parameter σ_*^2 .*

Proof. Take any subsequential continuous-path joint limit B_1, B_2 of (16) under $P_{0,0}^{++}$, on fixed horizons (enlarge a horizon if needed at endpoints), and let $\mathcal{G}_s$ be the joint past. By (18)

$$\int_0^T \mathbf{1}_{\{B_1(s)=B_2(s)\}} ds = 0 \quad \text{almost surely.} \quad (21)$$

One can integrate continuous cutoffs of the gap supported in shrinking neighborhoods of zero; their grid sums give the same limiting expectations by joint tightness with continuous limits.

Conditional marginal increments. At a deterministic time s , put $H_r = \lfloor sn(r) \rfloor$ and restart the prelimit pair at $\sigma(H_r)$. This is a stopping index of the chain of common words, not a stopping time for an individual walk. All first-hit coordinates of both paths at levels at most H_r are determined by the words traversed through this boundary. Its conditional suffix law is the fresh shared conditioned law from the terminal pair of sites.

By (6), $\sigma(H_r) - H_r$ is uniformly tight. Consequently its normalized time shift vanishes. Joint tightness with continuous limits, on a slightly enlarged horizon, makes the values at $H_r + \lfloor tn(r) \rfloor$ and at $\sigma(H_r) + \lfloor tn(r) \rfloor$ differ by $o_{\text{Pr}}(r)$ after subtracting the same linear centering. At the initial endpoints this also follows directly from (6); the centering correction is a tight integer times θ_r/r , which vanishes by Lemma 4.3.

Test against bounded continuous functions of finitely many joint past coordinates, represented at floor grid times in the prelimit. Condition at the common boundary and test the fresh normalized increment of length $\lfloor tn(r) \rfloor$. Lemma 4.4 makes its conditional bounded continuous test expectation converge to the normal value uniformly in the random terminal pair of sites. The time replacements just justified do not change the limit. Passing to the joint limit, then using a monotone-class argument for the joint past, proves that for each $t > 0$,

$$\mathcal{L}(B_i(s+t) - B_i(s) \mid \mathcal{G}_s) = N(0, \sigma_*^2 t), \quad i = 1, 2.$$

Off-diagonal cross increments. Moreover if g is a continuous $[0, 1]$ -valued cutoff vanishing on $[-2a, 2a]$, $a > 0$, and $\delta_i = B_i(s+t) - B_i(s)$, then for bounded $\mathcal{G}_s$ -measurable F ,

$$|E[Fg(B_2(s) - B_1(s))\delta_1\delta_2]| \leq \|F\|_\infty t e_a(t), \quad e_a(t) \longrightarrow 0 \quad (t \downarrow 0). \quad (22)$$

The bound can be uniform in s . To see this, again restart as above; on the supported past gaps the restart state has absolute gap $\geq ar$ except with vanishing error. Clip each future normalized increment symmetrically to the interval $[-w\sqrt{t}, w\sqrt{t}]$, for the moment fixed w, t . By (17) at the fresh state their product expectation differs from the independent-law one by at most $Cw^2 t \exp(-ca^2/t)$ in limsup; the independent-law value tends to zero by the single Brownian limits. Test first against continuous bounded past functions and pass to the joint limit. Removing the clipping there costs at most $t o_{w \rightarrow \infty}(1) \|F\|_\infty$, by the conditional marginal normals and Cauchy–Schwarz. Continuous past tests extend to bounded measurable ones, for instance by bounded approximation using $E(|\delta_1\delta_2| \mid \mathcal{G}_s) \leq \sigma_*^2 t$. Choosing $w \rightarrow \infty$ slowly gives (22).

Identification of the joint law. These facts force $B = (B_1, B_2)$ to consist of independent Brownian motions. For a direct verification fix s_0 , bounded F of the joint past there and a real vector $\xi \in \mathbb{R}^2$. Telescope the exponential of $i\xi \cdot (B(s) - B(s_0))$ on an equal mesh of $[s_0, v]$, multiply by F and take expectation. Taylor remainders through order two total $o(1)$ by the marginal Gaussian third absolute moment bounds. First-order terms vanish by the

past-conditional laws, and diagonal second-order terms give the Riemann sum using their variances. Cross terms total $o(1)$: away from gap zero by (22), and the complementary terms are bounded by a constant times the mesh sum of step lengths times expected continuous cutoff near gap zero (by the past-conditional absolute product bound), which is negligible by continuity and (21) as the cutoff shrinks. Thus $f(v) = E[F \exp(i\xi \cdot (B(v) - B(s_0)))]$ satisfies

$$f(v) = EF - \frac{1}{2}\sigma_*^2|\xi|^2 \int_{s_0}^v f(s) ds.$$

This gives the Gaussian increment law independent of the joint past, proving the assertion. $\square$

The passage from two copies to quenched concentration is a second-moment method developed in RWRE by Bolthausen and Sznitman [5, Lemma 4.1] and Berger and Zeitouni [4, Section 2]. The conclusion needed here is convergence in environment probability along Gaussian sequences; the following calculation supplies it under the present hypotheses.

Corollary 4.8 (Quenched Gaussian mass). *Along Gaussian sequences, the quenched path under D -conditioning from a fixed start has the Brownian limit of (16) in P -probability, in the sense of bounded continuous path tests. In particular, for every $\rho > 0$ and every $b_*(r) \rightarrow \infty$, with $H = \lfloor n(r) \rfloor$,*

$$P\left(\frac{K_H(0, \{W_H \leq b_*(r)r, |Y_H - b_H| \leq \rho r\})}{a_H} \geq c_\rho\right) \rightarrow 1 \quad (23)$$

for some $c_\rho > 0$ depending only on ρ and the walk law. The same assertion holds at translated starts.

Proof. Let F be a bounded real continuous test on path space, and let $M_r(\omega)$ be its expectation under the quenched conditioned path. Give the environment the probability law

$$\hat{P}(d\omega) = \frac{q(0)^2}{E[q(0)^2]} P(d\omega).$$

Under this weighting, two conditionally independent quenched conditioned paths have exactly the annealed law $P_{0,0}^{++}$. Proposition 4.7 and its marginal limits therefore give, with $b = E[F(B)]$ for the limiting Brownian path,

$$E_{\hat{P}} M_r \rightarrow b, \quad E_{\hat{P}} M_r^2 \rightarrow b^2.$$

Thus $M_r \rightarrow b$ in $\hat{P}$ -probability. Since $q(0) > 0$ almost surely, P is absolutely continuous with respect to $\hat{P}$; convergence in probability also holds under P .

For (23), use Lemma 4.3 to replace the linear centering by the medians. Choose a nonnegative continuous path test below the desired event, with a fixed bounded supremum and with the endpoint, throughout a small neighborhood of time one, strictly inside the window of radius ρ . Such a test can be chosen with positive Brownian expectation: the time-one centered normal law assigns positive mass to a smaller window, and continuity permits the nearby-time and sufficiently large fixed supremum restrictions. Eventually this fixed supremum bound is smaller than $b_*(r)$. The quenched no-drop-conditioned probability is consequently bounded below by a fixed positive constant with P -probability tending to one. Finally, on successful prefixes the total variation cost of replacing no-drop conditioning by conditioning on $T_H < T_{-1}$ is at most $1 - q(0)/a_H$, which tends to zero almost surely. This proves the stated normalization by a_H . Translation invariance gives the same conclusion at any translated start. $\square$

5 Clipping paths on arbitrary scales

The quenched limit in Section 4 applies only along Gaussian scales. We now obtain a weaker conclusion at every large scale: a fixed positive amount of quenched mass has its first-hit positions in an arbitrarily narrow tube around a suitable deterministic line. The line may depend on the scale, but not on the environment or the starting site.

Auxiliary paths and comparison with genuine quenched mass also appear in multiscale slowdown arguments; see Berger [2, Sections 5–6]. Here the retained mass is constructed from first-hit kernels using the spatial estimates of the preceding sections.

Proposition 5.1 (Clipping at arbitrary scales). *For any fixed $0 < T < \infty$, $\eta > 0$, $p_{\text{err}} > 0$ there is $d_0 > 0$ such that, for every sufficiently large r , some deterministic slope s_r satisfies, with $H_{\text{tot}} = \lceil Tn(r) \rceil$,*

$$P\left(K_{H_{\text{tot}}}(x, \{\max_{j \leq H_{\text{tot}}} |Y_j - js_r| \leq \eta r\}) \geq d_0\right) > 1 - p_{\text{err}}. \quad (24)$$

The bound holds at every prescribed starting site x , with the same d_0 and slope. It also gives the corresponding lower bound at every shorter horizon by taking first-hit prefixes.

The assertion is sitewise; it does not require one high-probability event on which all starting sites are good. This distinction permits the averaging over initial distributions in Section 6.

5.1 Transferring truncated moments

Lemma 5.2 (Truncated-moment transfer). *Write $\mathbb{Q}_h = E[K_h(0, \cdot)/a_h]$. For any $h \rightarrow \infty$, $z \rightarrow \infty$, put $X = W_h/z$ under $\mathbb{Q}_h$, and let $V = M_h^S/z$ have the iid sum law of (8). For fixed $0 < A < L < \infty$, with a constant independent of A, L ,*

$$\begin{aligned} \limsup \mathbb{Q}_h[(X \wedge L)^2] &\leq C \limsup \mathbb{E}[(V \wedge L)^2], \\ \limsup \mathbb{Q}_h[(X \wedge L)^2 \mathbf{1}_{X > A}] &\leq C \limsup \mathbb{E}[(V \wedge L)^2 \mathbf{1}_{V > A/C}]. \end{aligned} \quad (25)$$

Proof. Take a subsequence realizing the left lim sup and extract compactified weak limits μ, ν' in $[0, \infty]$ for the two laws. At each $u > 0$, $\mu([u, \infty])$ is bounded by the limit-law probability of the open event above $u/2$, hence by the corresponding prelimit lim inf. By (11) and (9), this is at most C_1 times a prelimit lim sup on the V side above $u/(2C_1)$; bounding by the limiting closed tail and enlarging it once more gives

$$\mu([u, \infty]) \leq C_1 \nu'((u/(4C_1), \infty]).$$

For the first inequality of (25) simply integrate with weight $2u$ up to L , since capped squares are bounded continuous. For the second, first upper bound the left using $X \geq A$ on the limiting law (upper semicontinuous test). This gives $A^2 \mu([A, \infty])$ plus the tail integral between A and L . Transfer each term to bound the sum by a constant times the ν' -expectation of the capped square on the open tail above $A/(4C_1)$. By lower semicontinuity that last expectation is bounded by the prelimit lim inf on our subsequence. Enlarging constants gives (25). $\square$

5.2 A path policy on short blocks

Proof of Proposition 5.1. We will concatenate a fixed number of short blocks. In each block we choose a probability law on a restricted set of successful prefixes, except on a rare failure event. The two requirements are different: the chosen endpoint displacements must have

small variance, and the restriction must retain a fixed positive amount of the original kernel mass. The first requirement keeps the auxiliary first-hit process in a narrow tube; the second will transfer that conclusion back to the quenched walk.

Work along an arbitrary sequence $r \rightarrow \infty$, and pass to a subsequence on which

$$F(a) = \lim_{r \rightarrow \infty} \frac{m(ar)}{m(r)} \quad (a > 0)$$

exists. This extraction is possible because the ratios are monotone in a , and changing a by a factor changes $m(ar)$ by at most its square. Thus they are equicontinuous on compact logarithmic intervals. In particular $F(1) = 1$, and the limit $F_0 = F(0+)$ exists in $[0, 1]$. It suffices to prove the clipping assertion on every such further subsequence with some fixed positive mass bound. Otherwise there would be a sequence of scales at which every deterministic slope fails even for mass thresholds tending to zero.

The common block rule. Fix a small tube width $b > 0$, and consider a block of integer height $h \leq tn(r)$, with $t > 0$ small. Its length will be chosen separately in the cases $F_0 = 0$ and $F_0 > 0$. At its current starting site x , put

$$N_b = \{W_h \leq br\}, \quad G = K_h(x, N_b), \quad f = G/a_h(x).$$

Short-block survival, Equation (12), gives a number $d_b \in (0, 1)$, depending on the fixed b, t , such that

$$P(G < d_b) \leq C_b t^2$$

for all sufficiently large r . On $\{G < d_b\}$, mark a failure and make an auxiliary jump to the block top with second coordinate displacement b_h and all other horizontal displacements zero. Otherwise sample a prefix from $K_h(x, \cdot \cap N_b)/G$. In the case $F_0 > 0$, we will sometimes further restrict this sampling to endpoints close to b_h .

The block test and every sampled prefix use only departure rows below the block top. Their endpoint and auxiliary choices therefore leave fresh iid layers above the next base. Repeating any one of these translated rules over equal blocks makes the second-coordinate endpoint displacements U_h iid under environment averaging and auxiliary sampling. Write $\lambda_h = \mathbb{E}U_h$. Both genuine and dummy endpoints satisfy

$$|U_h - b_h| \leq br, \quad |\lambda_h - b_h| \leq br.$$

The dummy jumps keep the auxiliary process defined after a failure; only trajectories without a failure will later contribute genuine quenched path mass.

We next choose the block length and, when useful, its further endpoint restriction. In both cases the goal is to make the variance accumulated over $O(n(r)/h)$ blocks small compared with r^2 .

Case $F_0 = 0$: the broad restriction has small variance. Take $h = \lfloor tn(r) \rfloor$ with $t > 0$ fixed and small, and use the common block rule without further restriction. The truncated-moment transfer and Equation (8) give

$$\limsup_{r \rightarrow \infty} \mathbb{Q}_h \left[\frac{W_h^2 \wedge (br)^2}{r^2} \right] \leq CtF(b). \quad (26)$$

The normalization in the block rule costs only a fixed factor in this moment estimate:

$$\mathbb{E}[(U_h - b_h)^2] \leq 4\mathbb{Q}_h[W_h^2 \wedge (br)^2].$$

Indeed, on $\{f \geq 1/2\}$, sampling the restriction costs at most a factor two relative to $K_h/a_h(x)$. On $\{f < 1/2\}$, use $(br)^2$ and

$$P(f < 1/2) \leq 2\mathbb{Q}_h(W_h > br).$$

Dummy jumps have zero centered displacement. Since there will be $O_T(1/t)$ blocks, Equation (26) makes their total variance at most $C_T F(b)r^2$ in the limit. This can be made as small as needed by choosing b small.

Case $F_0 > 0$: select a Gaussian subscale. Put $z = \varepsilon r$ and $h = \lfloor n(z) \rfloor$. Limits in this case are taken first as $r \rightarrow \infty$ along the chosen subsequence, and then as $\varepsilon \downarrow 0$. We have

$$\frac{h}{n(r)} \longrightarrow \frac{\varepsilon^2}{F(\varepsilon)}.$$

For every fixed $a > 0$,

$$\limsup_{\varepsilon \downarrow 0} \limsup_{r \rightarrow \infty} n(z) \Pr(|S| > az) = 0.$$

To see this, bound the expression by a constant depending on a times $(m(az) - m(az/2))/m(z)$; its iterated limit is zero because $F(a\varepsilon), F(a\varepsilon/2), F(\varepsilon) \rightarrow F_0 > 0$.

The quenched Gaussian estimate (23) consequently implies that, for each fixed $\rho, b > 0$, there is $\alpha_\rho > 0$ such that

$$\limsup_{\varepsilon \downarrow 0} \limsup_{r \rightarrow \infty} P((K_h/a_h)\{W_h \leq br, |Y_h - b_h| \leq \rho z\}) < \alpha_\rho = 0. \quad (27)$$

Here and below a block kernel without a stated start is at the origin. For completeness, a failure of this implication would give $\varepsilon_j \downarrow 0$ and sufficiently large r_j for which the first j large-jump tests at thresholds $a = 1, 1/2, \dots, 1/j$ are less than $1/j$, whereas the probability in (27) stays bounded away from zero. Taking $z_j = \varepsilon_j r_j \rightarrow \infty$ gives a Gaussian sequence, and $b/\varepsilon_j \rightarrow \infty$ is an allowed tube multiple in (23). This is a contradiction.

We also need a moment estimate for the fallback to the broad restriction:

$$\limsup_{\varepsilon \downarrow 0} \limsup_{r \rightarrow \infty} \frac{\mathbb{Q}_h[(W_h^2 \wedge (br)^2)\mathbf{1}_{\{W_h > Az\}}]}{z^2} \leq \frac{C}{A^2} + C \frac{F(b) - F_0}{F_0}. \quad (28)$$

Fix A, b, ε with $b > A\varepsilon$ and $b > \varepsilon$, and apply Equation (25) with $L = b/\varepsilon$. For the resulting iid sums, split each symmetric increment into its parts on

$$\{|S| \leq z\}, \quad \{z < |S| \leq br\}, \quad \{|S| > br\}.$$

All three parts are centered. The maximal sum U of the first parts has fourth moment at most Cz^4 , since $hm(z) \leq z^2$. The second maximal sum U' has second moment bounded by $Ch\mathbb{E}[S^2; z < |S| \leq br]$. The probability that any third part occurs, multiplied by the cap $(br)^2$, is bounded by $h(br)^2 \Pr(|S| > br)$. These last two bounds add to at most

$$Ch[m(br) - m(z) + z^2 \Pr(|S| > z)].$$

Without a third jump, the original maximal sum is at most $U + U'$. On the event that it exceeds Az , its capped square is bounded by a fixed constant times $U^2 \mathbf{1}_{\{U > Az/C'\}} + (U')^2$. Divide by z^2 , use the fourth-moment bound for the first term, and take the indicated iterated limits. The last large-jump term vanishes; this proves (28).

We now refine the common block rule. Let

$$g = (K_h/a_h(x))\{W_h \leq br, |Y_h - b_h| \leq \rho z\}.$$

On a nonfailure, when $g \geq \alpha_\rho$, sample this stricter restriction, normalized. When $g < \alpha_\rho$, use the broad restriction as before. Take $\alpha_\rho \leq 1$. The raw mass of whichever restriction is selected is then at least $\alpha_\rho d_b$: in the stricter case this follows from $a_h(x) \geq G \geq d_b$.

For $Az < br$, the endpoint displacement of this refined rule satisfies

$$\frac{\mathbb{E}[(U_h - b_h)^2]}{z^2} \leq \rho^2 + A^2 P(g < \alpha_\rho) + 4 \frac{\mathbb{Q}_h[(W_h^2 \wedge (br)^2) \mathbf{1}_{\{W_h > Az\}}]}{z^2}. \quad (29)$$

The stricter restriction contributes at most ρ^2 . In the fallback case with $f \geq 1/2$, split at $W_h = Az$ and use the factor-two normalization bound. When $f < 1/2$, use the same bound on its probability as in the first case, noting that $\{W_h > br\} \subset \{W_h > Az\}$. Again the dummy displacement costs zero. Equations (27)–(28) show that the iterated limit superior in (29) can be made arbitrarily small: first choose ρ, b small and A large, then choose ε small.

Concatenation and transfer to genuine path mass. In either case, run $J = \lceil H_{\text{tot}}/h \rceil$ blocks with the selected rule. In the second case set $t = 2\varepsilon^2/F_0$, which bounds $h/n(r)$ for large r . For small fixed ε , also $F(\varepsilon) \leq 2F_0$, so in both cases $J = O_T(1/t)$. The probability of any marked failure is at most

$$JC_b t^2 = O_{b,T}(t).$$

For the centered sums of the iid endpoint displacements, the maximal inequality for independent centered sums gives

$$\Pr \left(\max_{j \leq J} \left| \sum_{u=1}^j (U_h^{(u)} - \lambda_h) \right| \geq \eta r/2 \right) \leq \frac{4J\mathbb{E}[(U_h - b_h)^2]}{\eta^2 r^2}.$$

In the first case the limit superior of the numerator divided by r^2 is at most $C_T F(b)$. In the second case $Jz^2/r^2 \leq C_T$, and Equation (29) gives the same required smallness. Choose the parameters in the orders stated above. In the first case choose t sufficiently small after b ; in the second choose ε sufficiently small after ρ, b, A , with $t = 2\varepsilon^2/F_0$. In both cases require the failure probability to be small and

$$2b + C\sqrt{t} < \eta/2,$$

where C is the constant in Equation (10).

Within a genuine block, its median-tube restriction and Equation (10) imply

$$\max_{j \leq h} |Y_j - j\lambda_h/h| \leq (2b + C\sqrt{t})r < \eta r/2.$$

Thus, outside the failure and maximal-deviation events, the concatenated first-hit positions through Jh stay within ηr of the deterministic line of slope $s_r = \lambda_h/h$.

Choose the two annealed error bounds small enough that, with environment probability greater than $1 - p_{\text{err}}$, the conditional auxiliary law at the specified start assigns at least one half to these genuine tube paths. On a nonfailure, its density with respect to the local kernel is at most d_*^{-1} , where $d_* = d_b$ in the first case and $d_* = \alpha_\rho d_b$ in the second. The density of a genuine concatenation is therefore at most d_*^{-J} . The concatenated local kernels are submeasures of the original-floor kernel, by the quenched Markov property at first hits: each local no-drop condition merely adds a restriction. All parameters are now fixed, so $J \leq J_* < \infty$ for large r . The original kernel through Jh consequently gives the tube mass at least $d_*^{J_*}/2$. Taking its first-hit prefix at H_{tot} proves Equation (24). $\square$

6 Kernel tools

We need two ways to retain mass in a shared environment. The stage construction will require tuples of particles with separated endpoints and a bound on rapid loss of their mass. The stationary-profile argument will instead require a pair of particles whose signed gap does not decrease, at a cost growing only polynomially with the traversed height. We derive both conclusions from clipping and endpoint spread.

Given a probability π on k -tuples of sites in one layer, write

$$\Gamma_h^\pi = \pi K_h^{\otimes k}$$

using the same environment, as endpoint or prefix mass, and $\text{sep}(\mathbf{x}) = \min_{i \neq j} |(x_i)_2 - (x_j)_2|$. Write $\mathcal{R}_h(d') = \{\max_{s \leq h, i, j} |Y_s^{(i)} - Y_s^{(j)}| \leq d'\}$. The event $\mathcal{R}_h(d')$ concerns relative displacement differences at first hits, not absolute positions; s is the integer height offset from the indicated base. It reduces initial separation by at most d' . Such budgets add under concatenation at fresh layers. In each estimate under P , the initial law π is fixed independently of the upper environment. The same estimate therefore applies conditionally to a lower-layer-measurable initial law at a fresh layer.

We repeatedly use the following elementary composition rule. Retain a restricted submeasure of Γ_h^π , normalize its endpoint law, and continue with a local kernel from that law. The product of the two retained masses is then available in the longer original-floor kernel. This follows from the quenched Markov property at ordinary first hits, also for a product of walks: local no-drop conditions add restrictions to original-floor success. Taking prefixes similarly gives lower bounds at shorter horizons. Every finite-height mass, restriction, and normalization used below depends only on rows below its top; it can be computed by summing countably many finite path weights.

Lemma 6.1 (Clipping from an initial distribution). *For fixed $k, T, \eta, p' > 0$, with integer $k \geq 2$, there is a positive constant d such that, for every sufficiently large r , every $H \leq Tn(r)$, and every deterministic initial probability π ,*

$$P(\Gamma_H^\pi(\mathcal{R}_H(\eta r)) \geq d) > 1 - p'.$$

The analogous single-walk clipping bound holds from any initial probability, with the deterministic slope supplied by (24).

Proof. Apply (24) with tolerance $\eta/2$ and a sitewise failure probability ϵ . For every tuple, all k clipped kernels are good except with probability at most $k\epsilon$. Hence the expected π -fraction of bad tuples is at most $k\epsilon$, and with probability at least $1 - 2k\epsilon$ at least half of the tuple mass is good. On that event the product of clipped masses gives $d = d_0^k/2$. Choose $2k\epsilon < p'$. The single-walk assertion is the same argument with one coordinate. $\square$

When a shorter-horizon test is bounded using clipping through a longer horizon, the test itself still uses only the actual shorter kernel. All these uniform bounds apply conditionally to a past-measurable initial probability at a fresh layer.

6.1 Creating separated seeds

Proposition 6.2 (Separated seeds). *For each fixed $k \geq 2$, $p' > 0$, there are constants $C_*, c_0, g_0 > 0$ such that for all sufficiently large integer H , putting $r = r_{H/C_*}$,*

$$P(\Gamma_H^\pi\{\text{sep}(\text{endpoints}) \geq c_0 r\} \geq g_0) > 1 - p' \quad (30)$$

from arbitrary π , without initial separation.

Proof. Splitting one starting mass. First consider a single prescribed start. We will produce k ordered endpoint groups, with support gaps of order r , each carrying a fixed positive mass. Use J_0 blocks of length $h = \lfloor n(r) \rfloor$, followed by a remainder, to reach $H = C_* n(r)$, where J_0 is a large fixed integer and $C_* = 2J_0 + 2$. Clipping through $C_* n(r)$ supplies a common deterministic slope s_r and, with arbitrarily high environment probability, fixed positive mass within ηr of that line. The tolerance η will be small compared with $1/J_0$.

Endpoint spread, Equation (13), supplies a designated sign of a deviation of size at least cr from hs_r , with single-walk P^+ -probability at least $c > 0$. These constants are independent of the clipping parameters. The corresponding raw K_h -mass has expectation at least pc , by using paths on D , and is bounded by one. It therefore exceeds a fixed positive threshold on an environment event of fixed positive probability. The same reasoning applies after averaging over any starting probability.

Maintain the groups as submeasures of the walked mass, normalizing within each group for its next test. If there are fewer than k groups and the designated sign is positive, take the rightmost group and a second-coordinate median. Preserve its half at or below the median by central clipping. From the half at or above the median, retain the designated jumping mass whenever it exceeds its fixed threshold; these endpoints form a new rightmost group. Preserve all other groups by central clipping as well. For a negative designated sign, perform the reflected operation at the leftmost group.

Both median halves have mass fraction at least one half. A median atom may be used in both tests. In the positive-sign case, the retained central endpoints have an upper bound given by the median plus $hs_r + \eta r$, while the jumping endpoints have a lower bound given by the median plus $hs_r + cr$. The negative-sign case has the reflected inequalities. Thus the new groups have disjoint supports with a gap of at least $(c - \eta)r$. Every old gap loses at most $2\eta r$ under central propagation. Once there are k groups, only their central preservation is needed.

The initial-distribution clipping lemma makes all the central preservations succeed simultaneously with arbitrarily high conditional probability. Abort if a preservation fails. Subtracting this small error from the jump event shows that, whenever a split is still needed, preservation together with a split has conditional probability at least a fixed $s_* > 0$, independent of η, J_0 . The operations are measurable: take full restrictions to the indicated events and choose, for example, lower integer medians.

Choose J_0 large, and then choose the preservation errors small compared with $1/J_0$. The probability of no abort but fewer than $k - 1$ splits in these blocks is at most

$$\sum_{i < k-1} \binom{J_0}{i} (1 - s_*)^{J_0-i}.$$

Indeed, on such a pattern a split is needed at every step, and each prescribed nonsplit without abortion costs at most $1 - s_*$ by conditioning on the previous blocks. Apply central clipping once more over the remaining height, which is at most $C_* n(r)$. Taking η sufficiently small after fixing c, J_0 leaves final gaps at least $c'r > 0$. The product of the finitely many retained mass fractions gives a fixed positive lower bound for each group in the original single-walk kernel K_H . Every test and continuation uses only layers below its actual endpoint.

Choosing separated endpoints. For any starting tuple whose k starts each have these k options, the product kernel gives fixed positive mass to separated endpoint tuples. Choose $2c_0 < c'$ and expose endpoints successively. Every earlier endpoint excludes at most one group at the next start, since two distinct groups are separated by more than $2c_0 r$. Thus at the i th choice at least one of the k groups remains available. Each choice costs only its fixed positive group-mass lower bound. Averaging the favorable fraction over π , as in Lemma 6.1, proves (30). $\square$

6.2 Rapid loss bound

Fix $\lambda \in (0, 1]$ satisfying (4) and write

$$q_\pi = \int \prod_{i=1}^k q(x_i) \pi(d\mathbf{x}).$$

Lemma 6.3 (A separated inverse-moment bound). *For each fixed positive $B_* < \infty$ and integer k , if initial separation on the support of π is sufficiently large, depending on B_*, k , then*

$$P(q_\pi < 2s) \leq e^{Ck} s^\lambda, \quad s \geq e^{-B_*}, \quad (31)$$

with C independent of B_*, k .

Proof. Truncate each q upward to $q \vee e^{-B_*}$, changing the product by at most e^{-B_*} : if a factor changes, the entire new product is at most this quantity. By bounded product-field mixing at mutually separated sites, the expectation of the inverse λ -power of the truncated product is bounded by $2(Eq(0)^{-\lambda})^k$ for sufficiently large separation. To see the required uniformity, at fixed k, B_* approximate the bounded single-site functional in $L^1(P)$ by one using finitely many rows and having the same bound. Sufficient separation makes all its translates disjoint. On the event on the left of (31), the truncated product integral is $< 3s$. Jensen's inequality for its negative power and Markov's inequality give (31). $\square$

Proposition 6.4 (Rapid loss is unlikely). *Suppose π starts with $\text{sep} \geq G$ and $0 < d' \leq G/2$, with $G - d'$ sufficiently large for (31). For any positive integer L' and $kL' \log(1/\kappa) \leq x \leq B_*$, the loss $Z = -\log \Gamma_H^\pi(\mathcal{R}_H(d'))$ satisfies*

$$P(Z > x) \leq \left(e^{Ck - \lambda x/L'} + Ck e^{x/L'} H/n(d'/(2L')) \right)^{L'}. \quad (32)$$

Here C can be fixed independently of k, L', B_*, H, d' , given the stated separation condition.

Proof. From the current layer in a subdivision, monitor the local path mass with budget d'/L' , stopping at the first height where it is $< s := \kappa^{-k} e^{-x/L'}$. The assumptions give $e^{-B_*} \leq s \leq 1$. At any fresh start whose separation is still at least $G - d'$, the probability of such a threat is bounded by the parenthesis in (32). Indeed the raw mass of all k infinite D -successes is exactly the current q -product integral. Its part violating the budget within H layers has expectation at most $CkH/n(d'/(2L'))$, by (8)–(9) and a union over single walks on D with a median deviation greater than $d'/(2L')$. The other path factors are bounded by one and can be dropped. If the product integral is at least $2s$ and the violating mass at most s , enough mass satisfies the budget at every first hit to prevent a threat. Lemma 6.3 and Markov's inequality now give the claimed bound; the factor $\kappa^{-k\lambda}$ is absorbed into e^{Ck} . Neither D nor q is used to choose the stopping layer: the actual test is the finite-height mass, measurable from rows below that layer.

At a threat, extend the passing mass from one layer earlier by a direct upward step for every particle; at offset zero the passing mass is one. This preserves relative displacement errors and retains mass at least $\kappa^k s = e^{-x/L'}$. Normalize at the actual stopping layer and restart with the fresh upper environment. A final piece with no threat also has mass at least $e^{-x/L'}$.

If the horizon finishes within L' pieces, concatenation gives original mass at least e^{-x} , with total relative-motion budget at most d' . Otherwise each of the first L' pieces has detected a threat before completion. The conditional threat estimate applies at every restart: only lower rows have been revealed, and separation remains at least $G - d'$. Multiplying these conditional bounds gives Equation (32). $\square$

6.3 An outward-order kernel

For any same-layer sites x, y put

$$p_H^\rightarrow(x, y) = (K_H(x) \otimes K_H(y)) \{Y_H^{(2)} \geq Y_H^{(1)}\}.$$

The inequality concerns movement displacements; the two quenched paths use the same environment. Thus it asks that their signed physical gap not decrease from its initial value.

Proposition 6.5 (Outward order at polynomial cost). *There are fixed $A_\rightarrow, c > 0$ and a nonnegative random function $\mathcal{Z}(x, y)$ of the upper environment such that, simultaneously in integer $H \geq 1$,*

$$-\log p_H^\rightarrow(x, y) \leq A_\rightarrow \log(1 + H) + \mathcal{Z}(x, y), \quad \sup_{x, y} E \exp(c\mathcal{Z}(x, y)) < \infty. \quad (33)$$

Every cap $\mathcal{Z} \wedge B'$, $0 < B' < \infty$, is approximable in P -probability uniformly in x, y by bounded functions of fixed-radius upper row neighborhoods of these sites.

Proof. We construct one sequence of retained endpoint laws and use it at every target height. A stage starts with an extra signed gap, or buffer, between the walks. It usually increases this buffer by a fixed factor; on failure it reduces the buffer by a smaller factor. The resulting positive logarithmic growth limits the number of stages through height H to order $\log(1 + H)$, apart from an exponentially integrable random error. Each stage costs only a fixed amount of logarithmic mass.

Endpoint tests and retention. Keep the original signed gap $z_0 = y_2 - x_2$ fixed, even when it is negative. First prescribe $\lceil R_0 \rceil$ positive lateral steps for walk 2 and one upward step for each walk. This bounded uniformly elliptic script supplies a buffer R_0 , to be fixed large enough below, at a fixed positive mass cost.

At a stage base, the current pair probability π is supported on signed gaps at least $z_0 + R$, where $R \geq R_0$. Plan $h = \lfloor n(R) \rfloor$ layers. For constants $a > 0$, $0 < \ell' < 1$, and $g_* > 0$ to be chosen, test

$$\Gamma_j^\pi \{\text{signed endpoint gap} \geq z_0 + (1 - \ell')R\} \geq g_* \quad (1 \leq j \leq h).$$

Stop at the first failing test. If all pass, also test at h whether mass at least g_* has gap at least $z_0 + (1 + a)R$. These are tests of the full kernels from the same stage base. We do not intersect their endpoint events as path restrictions: a completed stage uses its retained final law, while a target height inside that stage uses only the test at that one height.

On success retain and normalize the growing endpoint mass, and replace R by $(1 + a)R$. On failure replace R by $(1 - \ell')R$. At a first traversal failure, take the passing mass from the preceding layer and extend each walk by one direct upward step. This preserves its signed gap and supplies mass at least $\kappa^2 g_*$ at the stopping layer; at offset zero the passing mass is one. If only the final growth test fails, its passing traversal test already supplies this much mass. Normalize the retained law. Whenever the new buffer is below R_0 , restore it by bounded positive lateral steps for walk 2 followed by an upward step for both walks. The buffer before restoration is at least $(1 - \ell')R_0$, so the script has uniformly bounded length.

Every test, retained law, and restoration uses only departure rows below its terminal layer. Conditional on this information, the next stage therefore starts in fresh iid layers.

A uniform chance of buffer growth. We choose a and a success probability lower bound $s_* > 0$ independently of small ℓ' . Clip both walks at scale R about the common deterministic slope in Equation (24), with tolerance δR , where $2\delta < \ell'$. The endpoint spread estimate (13), applied to the same deterministic center hs_R , gives one sign of a deviation of size at least $c_1 R$ with single-walk annealed P^+ -probability at least a fixed positive constant. The constants do not depend on the clipping parameters. The favorable sign may depend on R : use a positive deviation for walk 2, or a negative deviation for walk 1. Either choice increases their signed gap when the other walk stays near the center.

The raw jump mass has a fixed positive expectation, by restricting to D -paths, and is bounded by one. Thus it is at least some $j_0 > 0$ on an environment event of probability at least $c_0 > 0$, uniformly over starts and large R . Choose the sitewise clipping errors with sum less than $c_0/2$, and let $d > 0$ be their common retained mass bound. For each deterministic starting pair $\mathbf{x}$, the event $A_{\mathbf{x}}$ that this jump mass is at least j_0 and both clipped masses are at least d then satisfies

$$P(A_{\mathbf{x}}) \geq c_0/2.$$

This uses subtraction of the two clipping errors, with no independence between the events.

For an arbitrary current pair law, set $F = \int \mathbf{1}_{A_{\mathbf{x}}} \pi(d\mathbf{x})$. Since $EF \geq c_0/2$ and $0 \leq F \leq 1$,

$$P(F \geq c_0/4) \geq c_0/4.$$

On this event, central clipping for the favorable pairs supplies at least Fd^2 to every traversal test. At the terminal height, the favorable jump and the central restriction for the other walk supply at least Fj_0d to the growth test. The terminal paths need not be the paths used to prove the traversal tests; the tests were defined separately for this reason. Choose $\delta < c_1/2$, $a < c_1/2$, and

$$g_* \leq (c_0/4) \min(d^2, j_0d).$$

Then every stage succeeds with conditional probability at least a fixed $s_* > 0$, independently of small ℓ' . The constants g_* and R_0 may depend on the chosen ℓ' . All choices can be made deterministic at the countable set of buffer values reached by the procedure.

The number of stages through a given height. Fix ℓ' small enough that a proportion $s_*/2$ of successes gives positive logarithmic growth. Let S_m count successes in the first m main stages, and put

$$W = \sup_{m \geq 1} (s_* m/2 - S_m)_+.$$

The conditional success lower bound implies a uniform exponential tail for W . Indeed, for sufficiently small $\theta > 0$,

$$\exp\{\theta(s_* m/2 - S_m)\}$$

is a nonnegative supermartingale: the conditional multiplier is at most $e^{\theta s_*/2}(1 - s_* + s_* e^{-\theta}) \leq 1$. Its maximal inequality gives $P(W > u) \leq e^{-\theta u}$.

Success multiplies the buffer by $1 + a$, failure by $1 - \ell'$, and restoration only increases it. Consequently, after m completed main stages and their restorations,

$$\log R \geq c_2 m - C_2(W + 1)$$

for fixed positive constants c_2, C_2 . If H' is the height already consumed, we also have $R \leq C(H' + 1)$. A successful increase from R_{old} uses its full planned length $\lfloor n(R_{\text{old}}) \rfloor \geq cR_{\text{old}}$, by Equation (7) and a sufficiently large R_0 . Failures reduce the buffer, and restorations reset it to R_0 . Thus at any integer target height H , at most

$$C(\log(1 + H) + W + 1)$$

main stages have been completed, and at most one more is unfinished.

Concatenate the retained masses through H . If the last stage is unfinished, use its traversal test or its failure retention at the actual offset. If the target is the endpoint of a one-layer restoration, its bounded script supplies the continuation instead. The endpoint gap stays above the original baseline, so these paths contribute to $p_H^\rightarrow(x, y)$. Every completed stage, including a possible restoration, and the last partial stage cost a fixed logarithmic amount. Hence, simultaneously for all $H \geq 1$,

$$-\log p_H^\rightarrow(x, y) \leq C_3 \log(1 + H) + C_3(W + 1)$$

with a constant uniform in the starting pair.

An exponentially integrable local remainder. Take $A_\rightarrow = 2C_3$, and define

$$\mathcal{Z}(x, y) = \sup_{H \geq 1} [-\log p_H^\rightarrow(x, y) - A_\rightarrow \log(1 + H)]_+.$$

The preceding bound gives $\mathcal{Z} \leq C_3(W + 1)$, proving the exponential moment in Equation (33). It also gives

$$-\log p_H^\rightarrow(x, y) - A_\rightarrow \log(1 + H) \leq C_3(W + 1) - C_3 \log(1 + H).$$

On $\{W \leq M\}$, only a deterministic bounded range of heights can therefore contribute to the supremum. The probability of its complement tends to zero uniformly in x, y .

For a bounded height range, kernels can be uniformly approximated by bounded-length paths. Indeed, from any site inside a finite-height strip, a fixed number of consecutive upward steps forces exit and has uniformly positive probability. The strip exit time therefore has a geometric block bound, uniformly in the environment and horizontal start. Restricting both paths to a bounded number of steps changes their pair kernel mass by a uniformly small amount and uses only fixed-radius upper-row neighborhoods of the two starts. Those neighborhoods may overlap; the truncation estimate is unchanged. Finally,

$$p_H^\rightarrow(x, y) \geq \kappa^{2H},$$

by the two straight upward paths. Logarithms are therefore uniformly continuous on this bounded range of heights. Take the finite maximum, then cap at B' , and let the two truncation errors tend to zero. This proves the uniform local approximation of $\mathcal{Z} \wedge B'$. $\square$

7 Separated stages and episode bounds

Fix $\beta = 1/2$. Suppose (5) fails along integers $N \rightarrow \infty$ for some fixed D_0 , and let Q_N be the law of the environment conditioned on $a_N < N^{-\beta}$ along this sequence. Then

$$\text{KL}(Q_N \mid P) \leq D_0 \log N. \tag{34}$$

We will divide the interval of heights from zero to N into episodes. During an episode, a retained law of a fixed number of particles is repeatedly extended: a successful extension increases its separation scale, and a failed extension decreases that scale. An episode ends when the scale drops below a fixed floor. The construction below makes failures very unlikely under the fresh environment law P . We then use the displayed entropy bound to control their expected number under Q_N .

7.1 Separated extensions

Proposition 7.1 (Separated stages). *There are fixed constants $f, g, \chi, K > 0$ and an integer $k \geq 2$ with*

$$\frac{1}{g} < \frac{\beta}{64}, \quad \frac{(1 + f/g)D_0}{K} < \frac{\beta}{64},$$

having the following property. For every sufficiently large fixed b , put $B = kb$ and choose a sufficiently large fixed floor s_Π . At every level-count scale $s \geq s_\Pi$, put

$$s_- = se^{-fb}, \quad s_+ = se^{gb}, \quad H_e = \lfloor se^{-b} \rfloor, \quad H = \lfloor se^{\chi b} \rfloor.$$

For every probability π on k -tuples supported on $\{\text{sep} \geq r_s\}$, a fresh product environment satisfies

$$\begin{aligned} \Gamma_h^\pi \{\text{sep}(\text{endpoints}) \geq r_{s_-}\} &\geq e^{-B} \quad (1 \leq h \leq H), \\ \Gamma_H^\pi \{\text{sep}(\text{endpoints}) \geq r_{s_+}\} &\geq e^{-B} \end{aligned} \quad (35)$$

with probability at least $1 - e^{-2Kb}$. Each test uses the full kernel from the same stage base; a passing test does not restrict the paths used by later tests.

Proof. The main construction gives several chances to create separated mass before any prescribed short interval of target heights. A failed chance reveals only layers below the next available chance, so the estimate can be repeated there. We first describe these chances and their probability bounds, and then choose the constants to cover all target heights. Throughout these estimates $A \geq 1$ and the integer $k \geq 2$ are provisional fixed parameters. Write

$$\Lambda = \log 16, \quad j_* = \frac{1}{8A}, \quad m = \lfloor j_* b \rfloor.$$

We take b large enough that $m \geq 1$. All uses of the rapid-loss bound (32) will have cap $B_* = 2kb$; after fixing b , increasing s_Π will ensure its separation hypotheses.

A nested schedule before one target interval. Cover $[H_e, H]$ by integer cells $[v, v + w_m]$, where

$$w_0 = 16^m \left\lfloor \frac{H_e}{4 \cdot 16^m} \right\rfloor, \quad w_i = 16^{-i} w_0 \quad (0 \leq i \leq m),$$

and v ranges through $H_e, H_e + w_m, \dots$ up to H . For the growth test at H , add one cell with $v = H$, using H in place of H_e in the definition of w_0 . After increasing s_Π , each w_0 lies between one eighth and one quarter of its corresponding horizon. A certificate for the last cell may use paths above H . These certificates will only bound failure probabilities. The actual kernel tests in (35) alone determine the operational stopping layers.

Fix one cell. Its m opportunities have starting heights

$$v - 2w_i, \quad 0 \leq i < m.$$

All these heights are nonnegative and increase with i . When an attempt begins at opportunity i , use the normalized endpoint law of the full original stream $\Gamma_{v-2w_i}^\pi$. Uniform ellipticity makes this mass positive, and it uses only rows below $v - 2w_i$. In particular, after an earlier failure we restart from this full stream, not from the mass retained by the failed attempt.

The attempt first runs a seed kernel over w_i layers. Apply (30) with error e^{-6k} , and put

$$\rho_i = r_{w_i/C_*}.$$

Except with that error, it supplies mass at least g_0 whose endpoint separation is at least $c_0 \rho_i$, with $c_0 \leq 1$. On failure we consume opportunity i and try $i + 1$, if one remains. The failure is known at height $v - w_i$, which is below $v - 2w_{i+1}$.

On seed success, normalize its separated mass and protect it through the whole target cell. Number the protection steps $l = 0, \dots, m - i - 1$. Step zero starts at $v - w_i$; a nonfinal step l ends at $v - 2w_{i+l+1}$, and the final step ends at $v + w_m$. Every step has length at most $3w_i 16^{-l}$. Thus, if failure is detected through a nonfinal step l , opportunity $i + l + 1$ begins exactly at its detection height. The final step covers the entire cell and leaves no retry.

Retaining the separated seed. Choose an integer cutoff l_0 , sufficiently large as specified below. Combine the first $\min(l_0, m - i)$ protection steps into a single kernel, ending at the same height as that last combined step. Its length is at most $3w_i = 3C_* n(\rho_i)$. Apply Lemma 6.1 with

$$T = 3C_*, \quad \eta = c_0/4, \quad p' = e^{-6kl_0}.$$

Except with probability at most e^{-6kl_0} , this retains mass $g_1 > 0$ with relative-motion budget $c_0\rho_i/4$ throughout the combined protection. On success normalize this mass. On failure consume the $\min(l_0, m - i)$ opportunities and restart at the next opportunity if one remains. The test uses the kernel only through its actual endpoint, including when fewer than l_0 steps remain.

At every remaining step $l \geq l_0$, allow relative-motion budget

$$d_l = (c_0/16)\rho_i 2^{-l}.$$

Let Z_l be minus the logarithm of the mass respecting this budget, from the current normalized law. Continue the attempt if

$$\sum_{u=l_0}^l Z_u \leq Ak(l+1),$$

retaining and normalizing that budget-respecting mass. Otherwise declare failure at the endpoint of this step. The early budget is $c_0\rho_i/4$, and all late budgets together are at most $c_0\rho_i/8$. Consequently all current laws have separation at least $c_0\rho_i/2$. Every restriction and normalization uses only rows strictly below its endpoint. Figure 2 depicts the underlying height schedule.

Costs of the late protection steps. Set $a' = 2Ak$ and $\lambda' = \lambda/4$. We will obtain, conditionally at every active late step,

$$\begin{aligned} E[e^{\lambda'(Z_l \wedge a'(l+1))} \mid \text{past}] &\leq e^{C_{\text{mg}}k}, \\ P(Z_l > a'(l+1) \mid \text{past}) &\leq e^{-(\lambda/2)a'(l+1)}, \end{aligned} \tag{36}$$

where C_{mg} is independent of $A \geq 1$ and $k \geq 2$. The cutoff l_0 , unlike C_{mg} , may depend on these parameters, the seed constants, and fixed retry counts. The rapid-loss estimate is uniform over normalized starting laws with the required separation; it does not depend on the early retained mass g_1 .

Indeed, (7) gives, for each fixed retry count L' ,

$$\frac{3w_i 16^{-l}}{n(d_l/(2L'))} \leq 3C_* \left(\frac{32L'}{c_0} \right)^2 \left(\frac{4}{16} \right)^l =: C_{k,L'} e^{-\theta l}, \quad \theta = \log 4.$$

Choose a fixed $c_x > 0$ with $(1 + \lambda)c_x < \theta$ and $c_x < 1$. For $0 \leq x \leq c_x l$, use (32) with $L' = 1$. After increasing l_0 , its second term is at most $e^{-\lambda x}$. Below the applicability threshold $k \log(1/\kappa)$, use the trivial bound one. Both ranges therefore give

$$P(Z_l > x \mid \text{past}) \leq \min\{1, e^{C'k - \lambda x}\}, \quad 0 \leq x \leq c_x l,$$

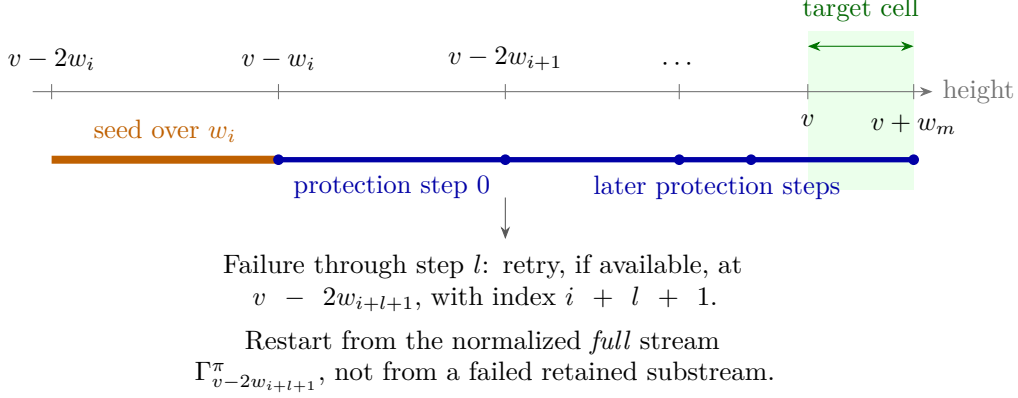


Figure 2: The height schedule of one opportunity, not sample paths. With $w_i = 16^{-i}w_0$, seeding runs from $v - 2w_i$ to $v - w_i$. Nonfinal protection step l ends at $v - 2w_{i+l+1}$; the final step ends at $v + w_m$ and covers the entire target cell. A failed nonfinal protection consumes the traversed opportunities, and any retry starts from the normalized full stream at its new base. There is no retry after the final opportunity. The marked first endpoint illustrates the geometric endpoint relation only. The initial $\min(l_0, m - i)$ protection steps are tested together at their last endpoint, so the diagram does not prescribe a retry at step zero. Heights are not drawn to scale.

with C' independent of A, k . For $c_x l \leq x \leq a'(l + 1)$, take a fixed integer L' sufficiently large that $\theta L' > (1 + 2\lambda)a'$, and increase l_0 again. The same rapid-loss estimate then gives

$$P(Z_l > x \mid \text{past}) \leq 2^{L'} e^{C_k L' - \lambda x} + (2C_k C_{k,L'})^{L'} e^{x - \theta l L'} \leq e^{-\lambda x/2}.$$

Here l_0 absorbs both the constants and the ellipticity threshold for this fixed retry count. Also $a'(l + 1) \leq 2Akm \leq kb/4 < B_*$. Finally, integrating these tails with $\lambda' = \lambda/4$ gives

$$\begin{aligned} E[e^{\lambda'(Z_l \wedge a'(l+1))} \mid \text{past}] &\leq 1 + \lambda' \int_0^\infty e^{\lambda' x} \min\{1, e^{C'k - \lambda x}\} dx + \lambda' \int_0^\infty e^{-\lambda x/4} dx \\ &\leq \frac{4}{3} e^{C'k/4} + 1 \leq e^{C_{\text{mg}}k}. \end{aligned}$$

This proves (36) with the required uniformity.

Why all opportunities rarely fail. Now impose $\lambda'A > C_{\text{mg}} + 8$. If the first cumulative late failure occurs at l , either $Z_l > a'(l + 1)$ or the sum of the truncated costs exceeds $Ak(l + 1)$. Earlier passing costs are below their own cutoffs, since they are nonnegative. Setting inactive costs to zero, conditional iteration of (36) bounds the second event by

$$\exp\{C_{\text{mg}}k(l - l_0 + 1) - \lambda'Ak(l + 1)\}.$$

Together with the cutoff tail, this is at most $e^{-5k(l+1)}$. A failed attempt consuming exactly n opportunities consequently has probability at most e^{-2kn} , measured at its start. The possibilities are seed failure ($n = 1$), early-protection failure ($n = \min(l_0, m - i)$), and a first late failure ($n = l + 1$); the stronger bounds just proved absorb any coincidence of these counts.

Each such failure is known before the next available opportunity, so revealing the full stream up to the new start leaves its upper layers fresh. Exhaustion partitions the m opportunities into positive consumed blocks. For a specified composition of m , conditional

iteration gives probability at most e^{-2km} . There are 2^{m-1} compositions, so the probability that this cell has no certificate is at most

$$e^{-(2k-\log 2)m}.$$

Choosing the constants. We can now choose constants in an order justified by the preceding estimates. Fix C_{mg} , then take $A \geq 1$ with $\lambda'A > C_{\text{mg}} + 8$, and define

$$f = 2 + \Lambda j_*, \quad \chi = g + 1 + \Lambda j_*.$$

Choose g, K with $1/g < \beta/64$ and $(1 + f/g)D_0/K < \beta/64$, and then choose $k \geq 2$ so large that

$$\lambda k/4 > 2K + 4, \quad (2k - \log 2)j_* > \chi + 2 + \Lambda j_* + 2K + 4. \quad (37)$$

Next fix the seed constants and retry counts, choose l_0 to validate (36), and then obtain g_1 from early clipping with its specified error e^{-6kl_0} . All these constants precede b . Thus even this very small early-protection error causes no circularity: its possibly small retained mass g_1 is fixed before we enlarge b . Finally enlarge s_{fl} , after b , to meet all scale and separation thresholds.

The full stream and the short heights. By (31), outside an event of probability at most $e^{-(2K+2)b}$ for sufficiently large b ,

$$\|\Gamma_h^\pi\| \geq q_\pi \geq e^{-B/4} \quad \text{for every } h.$$

We combine this exception with the cell failures by a union bound; none of the cell-attempt estimates is conditioned on this full-stream event.

To cover heights up to H_e , apply (32) at horizon H_e , with $d' = r_s/2$, $x = B/2$, and a fixed retry count $L' > k/2 + 2K + 4$. By (7),

$$\frac{H_e}{n(d'/(2L'))} \leq C(L')e^{-b}.$$

Raising the rapid-loss bound to its indicated power gives an upper bound of the form

$$C(e^{-\lambda kb/2} + e^{-(L'-k/2)b}) \leq e^{-(2K+2)b}$$

for large b , by (37). Off this event, taking prefixes gives mass at least $e^{-B/2}$ at every height through H_e with relative-motion budget $r_s/2$. Initial separation is at least r_s , so these prefixes have separation at least $r_s/2 \geq r_{s-}$, again by (7).

Assembling the cell certificates. The total number of cells, including the separate growth cell, is at most

$$C \exp[(\chi + 1 + \Lambda j_*)b] + 2 \leq \exp[(\chi + 2 + \Lambda j_*)b].$$

On successful completion of the attempt from opportunity i , separation throughout that cell is at least $c_0\rho_i/2$. The scale inequality gives

$$n(c_0\rho_i/2) \geq \frac{c_0^2}{4C_*} w_i.$$

For the traversal cells,

$$w_i \geq w_m \geq cse^{-(1+\Lambda j_*)b},$$

which exceeds the required scale $s_- = se^{-fb}$ after multiplication by the preceding fixed constant. For the growth cell,

$$w_i \geq cse^{(\chi - \Lambda_{j*})b},$$

which similarly exceeds $s_+ = se^{gb}$. In each comparison the remaining factor e^b absorbs the fixed constants for large b . Thus the certificate supplies the separation required in (35) at every height of its cell.

On the separately controlled event $q_\pi \geq e^{-B/4}$, concatenate the full stream to the attempt start with the restricted seed and protection kernels. Their mass, and therefore their prefix mass at each height of the cell, is at least

$$e^{-B/4}g_0g_1e^{-Akm} \geq e^{-B},$$

because $Akm \leq B/8$ and g_0, g_1 are fixed before b . The cell lies entirely in the protected part of the construction.

A union bound over cells, using (37), bounds the probability of any missing cell certificate by $e^{-(2K+2)b}$ for large b . Adding the full-stream and short-height exceptions proves (35). At fixed b , all the smallest scales tend uniformly to infinity with s_Π . Enlarging that floor therefore satisfies every scale and separation threshold used above. $\square$

7.2 Episode operation

We now use the stage estimate to construct the actual episode boundaries. Only the kernel tests in (35) determine those boundaries; the longer cell certificates in its proof do not enter the stopping rule. Every retained measure below is a deterministic restriction of a kernel, followed by normalization. No sample of a path is required.

Starting an episode. Start at height zero, and start a new episode after each reset below N . At an episode's initial height $t < N$, take the global normalized endpoint law

$$\mu_t = K_t(0, \cdot)/a_t$$

in product k times. Inject this tuple law into layer $t + 1$ with separation at least r_{s_Π} , using bounded elliptic scripts. Concretely, choose k fixed positive integer offsets in coordinate two, spaced by more than $2r_{s_\Pi}$. Prescribe one particle's endpoint at a time, choosing the first offset whose endpoint is at distance at least r_{s_Π} in coordinate two from every previously prescribed endpoint. Each earlier endpoint excludes at most one of the k choices, so a choice always exists. Move that particle laterally by its chosen offset and then directly up one layer.

The scripts have a fixed length bound and hence cost at most a fixed logarithmic mass B_0 . Normalize the retained tuple mass and set $s = s_\Pi$. If the injection reaches N , end the episode there without a stage.

Testing one stage. At the current height $u < N$, let π be the retained normalized tuple law and let $s \geq s_\Pi$ be its scale. Plan $H = \lfloor se^{\chi b} \rfloor$ layers. Test the traversal inequality in (35) successively at offsets

$$h = 1, \dots, \min(H, N - u).$$

Stop at the first failure. If offset H is reached, also test the growth inequality there. All these tests use the same initial law π and the full kernel from u ; passing an intermediate test does not restrict the mass used for later tests.

Retaining mass and updating the scale. If the full planned stage passes, retain and normalize the growing separated mass, and replace s by $s_+ = se^{gb}$. If a test fails, replace s by $s_- = se^{-fb}$ and retain mass with the corresponding smaller separation. For a failed growth test, the traversal test at the same height already supplies this mass. For a first traversal failure at offset h , extend the passing mass at $h - 1$ by one direct upward step for every particle. At $h = 1$ use the initial probability law as that passing mass. These upward steps preserve separation, and in every failure case the retained mass is at least

$$e^{-B} \kappa^k.$$

Normalize it. If the updated scale is below s_{ff} , end the episode and, if its endpoint is below N , start the next episode by the preceding injection rule. Otherwise continue with the next stage at the new scale.

At height N , end in all cases. A passing stage stopped at an offset strictly less than H is called incomplete; its traversal test supplies mass at least e^{-B} . There is at most one such stage. A failure encountered before this truncation still implies a failure of the untruncated tests in (35), so it has the same conditional probability bound under the fresh law P .

Every test and retention uses only rows strictly below its actual endpoint. Therefore all stage and episode boundaries are stopping times for the layer filtration. In particular, the upper layers at every subsequent stage start are fresh under P .

Let J, F be the total numbers of stages and failed stages, and let M be the number of episodes, with boundary heights $0 = t_0 < t_1 < \dots < t_M = N$. Put

$$h_i = t_{i+1} - t_i, \quad J_i = \#\{\text{stages in episode } i\}, \quad c_i = \log(a_{t_i}/a_{t_{i+1}}), \quad 0 \leq i < M.$$

Lemma 7.2 (Episode bounds). *For sufficiently large fixed b , the episode construction satisfies $M \leq J + 1$ and*

$$0 \leq c_i \leq B_0/k + 2bJ_i, \quad \log(1 + h_i) \leq C(1 + J_i). \quad (38)$$

For large N ,

$$J \leq \frac{\log N}{gb} + (1 + f/g)F + 1. \quad (39)$$

Under Q_N , fixed constants $c, C > 0$ and $L_0 < \infty$ satisfy

$$\begin{aligned} c \log N &\leq E_{Q_N} M, \quad E_{Q_N} (M + J) \leq C \log N, \\ E_{Q_N} \sum_{i < M} c_i \mathbf{1}_{\{J_i \leq L_0\}} &\geq c \log N. \end{aligned} \quad (40)$$

With the padding $t_i = N$ for $i \geq M$, each t_i is a bounded layer stopping time, and $\{i < M\}$ is known at t_i .

Proof. Mass and height within an episode. The unrestricted quenched k -survival mass through t_i is $a_{t_i}^k$. Starting from its normalized endpoint law, our retained continuation costs at most $B_0 + (kb + k \log(1/\kappa))J_i$ in logarithmic mass. Its concatenation is contained in the original survival kernel through t_{i+1} , so

$$a_{t_{i+1}}^k \geq a_{t_i}^k \exp\{-B_0 - (kb + k \log(1/\kappa))J_i\}.$$

This proves the upper bound for c_i in (38) once $b \geq \log(1/\kappa)$; its nonnegativity follows from the monotonicity of a_t . Each stage length is at most $se^{\chi b}$, and the scale increases from the fixed floor by at most a factor e^{gb} per stage. Summing these geometric bounds, together with the one-layer injection, proves the height bound in (38). Every episode except possibly the last contains a failed stage, so $M \leq J + 1$. The last episode may consist only of its injection.

Counting stages by their scale changes. For this accounting only, update the scale after the last completed or failed stage as if the operation continued, and clamp it upward to the floor after a reset. Every completed success consumes its full planned height $H \geq s_+$, so its updated scale is at most $\max(s_{\text{fl}}, N)$. Failures decrease the scale, and the clamping returns it only to s_{fl} . If S is the number of completed successes, their increases of gb , minus the losses of at most fb per failure, therefore satisfy

$$gbS - fbF \leq \log(N/s_{\text{fl}}) \leq \log N$$

for large N . There is at most one passing incomplete stage, so $J \leq S + F + 1$. This proves (39).

Transferring the failure bound to Q_N . Under P , conditional on the past at any actual stage start, its failure probability is at most e^{-2Kb} . If I is its failure indicator, then

$$E_P[e^{KbI - e^{-Kb}} \mid \text{stage past}] \leq e^{-e^{-Kb}}(1 + e^{-2Kb}(e^{Kb} - 1)) \leq 1.$$

There are at most N actual stages, since every stage advances at least one layer. Pad the list to N slots, with active indicator A_r , and put $I_r = 0$ in inactive slots. Before each slot its activity is known, and

$$E_P[e^{KbI_r - e^{-Kb}A_r} \mid \text{past}] \leq 1.$$

The inactive factor is one. Conditional iteration, using $\sum_r I_r = F$ and $\sum_r A_r = J$, gives

$$E_P e^{KbF - e^{-Kb}J} \leq 1.$$

Applying the entropy inequality with (34) yields

$$KbE_{Q_N}F \leq D_0 \log N + e^{-Kb}E_{Q_N}J.$$

Only the exponential estimate was taken under the fresh law; this last inequality requires no independence under Q_N .

Write $c_f = 1 + f/g$. Combining the last inequality with (39) gives

$$E_{Q_N}J \leq \frac{\log N}{b} \left(\frac{1}{g} + \frac{c_f D_0}{K} \right) + \frac{c_f e^{-Kb}}{Kb} E_{Q_N}J + 1.$$

The coefficient in parentheses is less than $\beta/32$. After absorbing the exponentially small term, for large fixed b and then large N ,

$$2bE_{Q_N}J \leq \frac{\beta}{4} \log N.$$

Positive loss in episodes with bounded stage count. Under Q_N , $\sum_{i < M} c_i = -\log a_N > \beta \log N$. Summing (38) and using the last bound shows

$$\frac{B_0}{k} E_{Q_N}M \geq \beta \log N - 2bE_{Q_N}J \geq \frac{3\beta}{4} \log N.$$

This proves the lower bound on $E_{Q_N}M$; its upper bound, together with that on $E_{Q_N}J$, follows from $M \leq J + 1$. For any $L_0 > 0$, the loss in the remaining episodes satisfies

$$\sum_{i < M} c_i \mathbf{1}_{\{J_i > L_0\}} \leq \left(\frac{B_0}{kL_0} + 2b \right) J.$$

Although the injection cost B_0 may be large, it is fixed. Choose L_0 after it so that the first term's expected contribution is at most $(\beta/4) \log N$. Episodes with more than L_0 stages then account for at most $(\beta/2) \log N$ of expected loss. Subtracting this from the total proves the last assertion of (40).

Finally, the operational tests use only rows below their tested layer, so all episode boundaries are bounded layer stopping times. With the stated padding, $\{i < M\} = \{t_i < N\}$ is known at t_i . $\square$

8 Stationary profiles and the contradiction

We retain the bad-event laws Q_N and the adapted episodes from Section 7. Their entropy and drop bounds (34), (38), and (40) will produce a stationary sequence of local endpoint profiles with positive mean normalization loss. We begin by identifying the statistic that will make this stationary law impossible.

For a full-support subprobability w on horizontal space, define

$$T_R(w) = \max_{j \in \mathbb{Z}} \min\{w(z_2 \leq j), w(z_2 \geq j + R)\}, \quad R \in \mathbb{Z}_{\geq 1}.$$

Here z_2 denotes the second full-space coordinate. This positive, translation-invariant quantity measures how much mass can be placed in each of two tails separated by R . The profile update multiplies surviving mass by the episode normalization factors. The outward-order estimate (33) transports the two tails at a cost logarithmic in the traversed height, together with a local environmental remainder. For a stationary marked profile law, we will derive

$$mEc_0 \leq A_{\rightarrow} E \log \left(1 + \sum_{i=0}^{m-1} h_i \right) + O(1),$$

with the last term independent of m . The logarithmic height moment from the episode bounds makes the right side $o(m)$, contradicting $Ec_0 > 0$.

The construction must supply a profile to which this argument applies. Local limits can initially have diffuse mass or several components escaping from one another. The array below retains those possibilities, then finite-window entropy excludes diffuse mass and reduces the positive-loss event to finitely many persistent components of full support. Marking one component uniformly preserves stationarity. The same entropy bound controls its upper environment relative to an iid field while retaining exactly its current profile and offset data.

Translation-invariant compactifications that retain separated components and diffuse mass were developed by Mukherjee and Varadhan [14, Theorem 3.2] for occupation measures and by Bates and Chatterjee [1, Theorem 2.8] for polymer endpoint distributions. We construct the array needed here directly, storing sampled-anchor offsets together with episode marks and the corresponding upper environments.

To remove the environmental bias from the local remainder, we first fix a finite number m of episodes and send the tail separation R to infinity. Only afterwards do we send m to infinity. No mixing estimate uniform in the episode window is needed. The final profile argument establishes the displayed obstruction in this order.

8.1 The stationary array

Horizontal space is $\mathbb{Z}^{d-1}$. We observe profiles from sampled anchors, allowing offsets between anchors to escape to infinity in the limit.

Pad each episode sequence to all integer indices by $t_i = 0$ for $i < 0$, $t_i = N$ for $i \geq M$, and zero marks h_i, J_i, c_i outside $0 \leq i < M$. Write $\mu_i^* = \mu_{t_i}$ for the normalized global endpoint profile there, viewed as a probability on horizontal sites. Conditional on the entire environment, sample anchors $U_{i,a} \sim \mu_i^*$ independently for indices $i \in \mathbb{Z}$ and $a \geq 1$. Keep the following array in a countable compact product:

- The nonnegative integer marks h_i, J_i and nonnegative real marks c_i , each range compactified by $+\infty$.
- The offsets $O_{i,a,jb} = U_{j,b} - U_{i,a}$, in the lattice with its one-point compactification, and all profile coordinates $w_j^{ia}(z) = \mu_j^*(U_{i,a} + z)$.

- The upper environments $\Omega^{i,m,a}(l, z)$, for $m \geq 1, l \geq 0$, and horizontal z . For $l < t_{i+m} - t_i$, use the true row $\omega((t_i + l, U_{i,a} + z), \cdot)$. At the remaining heights use the row at $(l, U_{i,a} + z)$ of a full auxiliary iid upper field for the pair (i, m) . This field is common to all anchors a , and the auxiliary fields are independent across pairs and independent of the environment and anchors.

Denote by $P_{\uparrow}$ the iid upper-field law on indices (l, z) , including $l = 0$. The functions $q(0; \Omega)$ and $a_H(0; \Omega)$ are the earlier quenched no-drop functions evaluated in this upper field; they require no rows below its base.

Profile and row coordinates use the ordinary compact ranges $[0, 1]$ and the row simplex, respectively. Let τ shift the array by reading all absolute episode indices one index later (leaving relative offsets in time such as m unchanged). Take the probability law $\mathbb{L}_N$ averaging shifts to episode starts:

$$\mathbb{L}_N[F] = (E_{Q_N} M)^{-1} E_{Q_N, \text{aux}} \sum_{i=0}^{M-1} F(\tau^i \text{array}).$$

Lemma 8.1 (Stationary episode limit). *Along a subsequence, $\mathbb{L}_N$ converges weakly to a shift-invariant law $\mathbb{L}$. Under this law, simultaneously for every i ,*

$$\begin{aligned} 1 &\leq h_i < \infty, & J_i, c_i &< \infty, \\ E_{\mathbb{L}}(c_i + \log(1 + h_i)) &< \infty, & E_{\mathbb{L}} c_i &> 0. \end{aligned}$$

Every stored row is uniformly elliptic with the original bound κ .

Proof. Compactness gives a subsequential weak limit. The shift is continuous, and the occupation sum telescopes: for every bounded continuous F ,

$$|\mathbb{L}_N[F \circ \tau] - \mathbb{L}_N[F]| \leq \frac{2\|F\|_{\infty}}{E_{Q_N} M} \longrightarrow 0.$$

At the origin, (40) bounds the occupation mean of J_i . Truncation passes this bound to the limit, so $J_0 < \infty$ almost surely. The bounds (38) then pass and give the stated integrability and finiteness. For the positive truncated-drop bound in (40), put $C_{L_0} = B_0/k + 2bL_0$ and use the globally bounded continuous test

$$(c_0 \wedge C_{L_0}) \mathbf{1}_{\{J_0 \leq L_0\}}.$$

It agrees with the required drop on every supported array, by (38). Its expectation therefore passes to the limit and gives $E_{\mathbb{L}} c_0 > 0$. The occupation law gives $h_0 \geq 1$, and shift invariance gives all assertions at every integer index. Uniform ellipticity is a closed condition on the row coordinates and therefore also passes to the limit. $\square$

Write

$$H_{i,m} = \sum_{s=i}^{i+m-1} h_s, \quad C_{i,m} = \sum_{s=i}^{i+m-1} c_s.$$

By the stationary ergodic theorem, in its possibly nonergodic form [7, Theorem 6.2.1], there exists $\zeta > 0$ and a shift-invariant positive-probability event

$$\mathcal{E} = \left\{ \lim_{m \rightarrow \infty} C_{0,m}/m > \zeta \right\} \tag{41}$$

(up to null sets).

8.2 Entropy in a finite episode window

We use standard relative-entropy tools originating with Kullback and Leibler [13]; the binary-event and projection inequalities are instances of data processing, as discussed in Sason and Verdú [17, Section II-B]. Their needed forms, including the variational formula, are proved below before being applied to the stopped episode windows.

Current data for entropy purposes are only

$$\mathcal{A}_i = ((O_{ia,ib})_{a,b}, (w_i^{ia}(z))_{a,z}).$$

The reference law keeps the marginal of $\mathcal{A}_i$. Conditionally on these data, labels at finite offsets share one iid upper field, translated by their offsets, while distinct classes use independent iid fields. The finite-offset relations in the limit are consistent equivalence relations with additive offsets: each specified finite-offset event is clopen, so every such consistency identity passes to the limit. Write $\mathbf{R}(\mathcal{A}_i, d\phi)$ for this reference conditional field law.

Lemma 8.2 (Finite-window entropy). *There is a constant $C_E < \infty$ such that, for every fixed i and $m \geq 1$, the $\mathbb{L}$ -law of $(\mathcal{A}_i, (\Omega^{i,m,a})_a)$ has relative entropy at most $C_E m$ with respect to its own current-data marginal followed by the reference kernel $\mathbf{R}$.*

Proof. We first record the elementary entropy facts used here. Relative entropy is $\text{KL}(Q | P') = \int f_0 \log f_0 dP'$, where $f_0 = dQ/dP'$, and is infinite without absolute continuity. It has the variational formula

$$\text{KL}(Q | P') = \sup_{F \text{ bounded measurable}} \left\{ QF - \log P' e^F \right\}.$$

The upper bound follows from Jensen applied to e^F/f_0 . For equality, truncate $\log f_0$ between $\pm B$ and send $B \rightarrow \infty$, or test a singular set. Thus projection cannot increase entropy. On compact metric spaces, bounded continuous tests suffice by bounded approximation in $L^1(Q + P')$; the variational formula therefore gives lower semicontinuity under joint weak convergence of both laws.

For an event of respective probabilities s, t , where $0 < t < 1$, projection to its indicator gives $\text{KL}(Q | P') \geq s \log(1/t) - \log 2$. If a reference law is a data marginal times an independent fixed law, replacing that marginal by the controlled law's *own* data marginal only decreases finite entropy. Indeed the new density is the old density divided by its conditional mean given the data, and the entropy difference is the nonnegative marginal entropy. These log-density identities are taken on sets of positive controlled density; negative parts are integrable, and conditional Jensen for $f_0 \log f_0$ justifies the marginal term.

Before the limit, at deterministic $i \geq 0$, put $\mathcal{H}_i = \sigma(\mathcal{F}_{t_i}, (U_{i,a})_a)$ and $\Phi_i = (\Omega^{i,m,a})_a$; $\mathcal{A}_i$ is $\mathcal{H}_i$ -measurable. Include the same auxiliary sampling under P . Given $\mathcal{H}_i$, the spliced field above t_i , in absolute horizontal coordinates, is still iid under P . Indeed, at both bounded layer stopping times the layers at and above the stopping layer are fresh, while the current anchors use only lower-layer data. Keeping the slab up to the second stopping layer and replacing its tail by an independent copy therefore preserves the iid law. Thus Φ_i has conditional law $\mathbf{R}(\mathcal{A}_i, d\phi)$.

Let $L_t = dQ_N|_{\mathcal{F}_t}/dP|_{\mathcal{F}_t}$. Under Q_N , the conditional density of the information stopped at t_{i+m} , together with the independent extension, relative to the corresponding P -law given $\mathcal{H}_i$, is $L_{t_{i+m}}/L_{t_i}$. The anchors at i use the same lower-data sampling kernel under both laws, and the extension costs no density. Conditional Jensen therefore gives, for every bounded F ,

$$\begin{aligned} E_{Q_N, \text{aux}}[F(\mathcal{A}_i, \Phi_i) | \mathcal{H}_i] \\ \leq E_{Q_N, \text{aux}}[\log L_{t_{i+m}} - \log L_{t_i} | \mathcal{H}_i] + \log \int e^{F(\mathcal{A}_i, \phi)} \mathbf{R}(\mathcal{A}_i, d\phi). \end{aligned}$$

Multiply by $\mathbf{1}_{\{i < M\}}$, which is past-known, and average as for $\mathbb{L}_N$. Jensen to pull the log outside this average gives the $\mathbb{L}_N$ entropy bound at index zero (against its own current-data marginal and the kernel) of

$$\frac{1}{E_{Q_N} M} \sum_{i \geq 0} E_{Q_N} [\log L_{t_{i+m}} - \log L_{t_i}] \leq \frac{m D_0 \log N}{E_{Q_N} M}.$$

Increments from $i \geq M$ are zero. The displayed sum telescopes over deterministic indices, since $t_i = N$ for $i \geq N$; its subtracted initial expected log densities are nonnegative relative entropies. For passage to the limit, finite-class consistency is a closed condition: self-offsets vanish, finite offsets are symmetric with sign change, and specified finite offsets add on triples. Violations are detected by finitely many clopen coordinate events. On this support the reference kernel is weakly continuous. For any finitely many field coordinates, coincidence of the input sites is specified by a finite-offset equality, while different sites have independent rows. Finite-coordinate tests suffice by compactness. Consequently the reference laws converge weakly with their current-data marginals. Lower semicontinuity and (40) yield the bound $C_E m$; shift invariance gives every i . $\square$

8.3 Profiles and multiplicities

Offsets across all times determine equivalence classes of labels (i, a) by finite distance. Within one class, the profiles at any time j , viewed from its different anchors, are translates of one another and have total mass at most one. These consistency statements pass coordinatewise.

Lemma 8.3 (Sampling identities). *Under $\mathbb{L}$, simultaneously for all indices and horizontal z ,*

$$2^{-n} \sum_{b \leq 2^n} \mathbf{1}_{\{O_{ia,jb}=z\}} \longrightarrow w_j^{ia}(z) \quad a.s. \quad (42)$$

A distinct label $(j, b) \neq (i, a)$ cannot have finite offset z from (i, a) while $w_j^{ia}(z) = 0$.

Proof. Before the limit, the anchor draws at j , excluding a possible own label, are conditionally independent samples given the environment and $U_{i,a}$. The occupation indices and weights depend only on the environment, not on these draws. The mean-square error in (42) is therefore $O(2^{-n})$, including the possible own-label contribution. The finite-offset indicators are continuous, so these summable bounds pass to the limit. Borel–Cantelli proves (42).

For distinct labels and any continuous function g on $[0, 1]$, the same conditional sampling identity gives

$$E_{\mathbb{L}} [g(w_j^{ia}(z)) (\mathbf{1}_{\{O_{ia,jb}=z\}} - w_j^{ia}(z))] = 0.$$

Approximate the indicator of zero by bounded continuous functions to obtain the second assertion. All the indices and sites are countable, so the assertions hold on one common probability-one event. $\square$

Lemma 8.4 (Profile updates and averaged survival). *For every i, m, a , write $H = H_{i,m}$, $C = C_{i,m}$, and $\Omega = \Omega^{i,m,a}$. Coordinatewise,*

$$w_{i+m}^{ia} \geq e^C w_i^{ia} K_H^\Omega. \quad (43)$$

Here the kernel maps horizontal starting positions at layer zero to endpoints at layer H in the anchor frame. Also, almost surely,

$$\limsup_n 2^{-n} \sum_{a \leq 2^n} a_{H_{i,m}}(0; \Omega^{i,m,a}) \leq e^{-C_{i,m}}. \quad (44)$$

Proof. Before the limit, local no-drop continuation is contained in the global survival stream. Dividing its endpoint inequality by the new global mass gives (43), with $C = \log(a_{t_i}/a_{t_{i+m}})$. At finite limiting marks, pass first the inequalities involving finitely many starting sites and bounded-length successful kernel paths. Strict violations of these finite-path inequalities are open, since each path weight is a finite product of row coordinates. Increasing the finite restrictions gives (43). Successful paths to layer H use no departure row at or above H , so the auxiliary tail is irrelevant.

Before the limit, at a fixed index the average in (44) has conditional mean at most e^{-C} given the environment, again by the global mass inequality. Its terms are independent bounded functions of the independently sampled current anchors; they do not use appended rows. Chebyshev bounds the conditional probability of exceeding $e^{-C} + \epsilon$ by $2^{-n}\epsilon^{-2}$, and this also holds after occupation averaging. For fixed finite H , the function a_H is lower semicontinuous, being a sum of nonnegative continuous finite-path weights. Thus the strict upper-violation event is open at finite marks, and the same summable probability bound passes to the limit. Borel–Cantelli, followed by a countable sequence $\epsilon \downarrow 0$, proves (44). $\square$

Call a current label *proper* if $w_i^{ia}(0) > 0$, and *dust* otherwise.

Proposition 8.5 (Finitely many persistent profiles). *On $\mathcal{E}$, there is no dust. The proper classes form the same finite nonempty set at every time, and the profile of each class has full support on horizontal space.*

Proof. *Dust is isolated.* If a dust label (i, a) had a distinct finite-offset neighbor (j, b) , with offset z , profile consistency would give $w_i^{jb}(-z) = w_i^{ia}(0) = 0$. Apply the second assertion of Lemma 8.3 with (j, b) as the reference label and (i, a) as the sampled target. This is impossible. Thus dust labels are singleton classes even when all times are included.

At any fixed time, the dust indicators are exchangeable, because anchor-permutation symmetry passes to the limit. They have an asymptotic frequency δ_i , which is positive if any dust occurs. To see the latter assertion, apply the ergodic theorem to the stationary label sequence of dust indicators. On the invariant zero-frequency event, taking expectations shows that each indicator vanishes.

Fix i, m . Under the reference law of Lemma 8.2, conditional on $\mathcal{A}_i$, the singleton dust classes have independent upper fields. Their values $q(0; \Omega^{i,m,a})$ are therefore iid with mean p . Finite entropy gives absolute continuity, so the strong law on the dust subsequence also holds under $\mathbb{L}$ whenever that subsequence is infinite. Since $a_H \geq q$, including in every spliced field, (44) yields

$$e^{-C_{i,m}} \geq p\delta_i.$$

This holds simultaneously for every m . On $\mathcal{E}$, its left side tends to zero, by (41) and stationarity. Dust is therefore absent at all times on $\mathcal{E}$.

There are finitely many proper classes. For fixed n and $r_0 > 0$, whenever possible choose from $\mathcal{A}_i$ labels in n distinct classes with self masses at least r_0 . On this selection event, when $C_{i,m} > \zeta m$, (43) and the total-mass bound for each output profile give

$$\prod_{\text{chosen}} q(0; \Omega^{i,m,a}) \leq r_0^{-n} e^{-n\zeta m}.$$

The event of successful selection together with this deterministic product bound is measurable from the current data and the extension fields. Its reference probability is at most

$$r_0^{-n\lambda} (Eq^{-\lambda})^n e^{-\lambda n \zeta m},$$

by (4) and class independence. The binary entropy bound and Lemma 8.2 show that its $\mathbb{L}$ -probability has $\limsup_{m \rightarrow \infty}$ at most $C_E/(\lambda n \zeta)$. On the selection event intersected with

$\mathcal{E}$, the product bound holds for all sufficiently large m . This bounds the probability of that intersection by the same quantity. Send $r_0 \downarrow 0$, then $n \rightarrow \infty$, to exclude infinitely many proper classes.

The classes persist and have full support. Over any positive integer height, the local kernel has positive weight between any two horizontal sites: move horizontally at the starting layer and then ascend directly. Uniform ellipticity makes this finite script positive. Thus (43) gives a full-support next profile for every current proper class. By (42), every positive coordinate is represented by next-time labels. Distinct classes cannot merge, by the global finite-offset equivalence relation.

Conditional on the shift-invariant event $\mathcal{E}$, the current number of classes is therefore nondecreasing, finite, positive, and stationary. It is consequently constant almost surely. Indeed, writing N_i for this count, stationarity and monotonicity give

$$E[\min(N_{i+1}, K) - \min(N_i, K)] = 0$$

for every integer K . The nonnegative difference vanishes almost surely; letting K increase proves $N_{i+1} = N_i$. This uses no moment of the number of classes. The class sets transport bijectively through time, and every profile has full support by the update from the preceding time. $\square$

8.4 Why a stationary profile cannot sustain normalization growth

The preceding construction has left us with a finite, persistent set of full-support profiles on the event $\mathcal{E}$. Condition on $\mathcal{E}$, mark one class uniformly, and use its least current label as the origin at each episode. Uniform marking commutes with time shift, so the marked law $\mathbb{L}_*$ is stationary. This choice matters: a class sampled according to its mass would generally receive a different weight after an episode.

For any fixed window of m episodes, put

$$H_m = \sum_{i=0}^{m-1} h_i, \quad C_m = \sum_{i=0}^{m-1} c_i.$$

Write w_0 for the marked profile at the start of this window and write w_m in the same horizontal frame. Both are subprobability measures of full support. The profile update gives

$$w_m \geq e^{C_m} w_0 K_{H_m}^{\Omega_m} \quad \text{coordinatewise,}$$

where Ω_m is the spliced upper field viewed from the current representative. The profile at time m , with its usual representative, has the same distribution as w_0 up to translation.

There is one further property of this marked field that we will need. Let $D_* = (\mathcal{A}_0, a)$, where $\mathcal{A}_0$ is exactly the current offset and profile data defined above and a is the least current label of the marked class. In particular, D_* contains no future episode marks or future profiles, and w_0 is a function of D_* . The joint law of (D_*, Ω_m) has finite relative entropy with respect to

$$\mathbb{L}_*(D_*) \otimes P_\uparrow. \tag{45}$$

Indeed, conditioning on $\mathcal{E}$ changes the density of the projected unmarked law by a factor at most $1/\mathbb{L}(\mathcal{E})$. Finite entropy therefore remains finite. Given the current data, uniform selection among the current class representatives is a probability kernel. Apply that same kernel to the reference law in Lemma 8.2; the selected field has iid law $P_\uparrow$ independently of the selected index and the current data. Projection and the entropy chain rule now give (45) with the controlled law's own data marginal.

The following argument isolates the remaining obstruction. It requires finite entropy separately for each fixed m ; no entropy bound uniform in m is used.

Proposition 8.6 (No positive stationary normalization rate). *Suppose a stationary marked profile law has full-support subprobability profiles and nonnegative marks c_i , positive integer marks h_i , with*

$$E[c_0 + \log(1 + h_0)] < \infty.$$

Assume the coordinatewise update (43), the finite entropy property (45) for every fixed m , and the outward-order estimate (33), including its uniform local approximation property. Then $Ec_0 = 0$.

Proof. We use the two-tail statistic introduced at the start of this section. For an integer $R \geq 1$, recall that

$$T_R(w) = \max_{j \in \mathbb{Z}} \min\{w(z_2 \leq j), w(z_2 \geq j + R)\}.$$

Here z_2 means the second full-space coordinate of a horizontal site. Full support and finite mass imply $0 < T_R(w) \leq 1$, and a maximizing integer j_R exists. Choose one measurably. For each fixed full-support w , its maximizing cuts satisfy

$$j_R \longrightarrow -\infty, \quad j_R + R \longrightarrow +\infty.$$

For example, if j_R were bounded below along a subsequence, the left mass at both j_R and $j_R - 1$ would stay bounded below whereas the right mass would tend to zero. Moving the cut one unit left would strictly increase the smaller mass, contradicting maximality. The other endpoint is treated by moving the cut right. Thus the normalized restrictions α_- and α_+ of w_0 to these two tails eventually sample sites outside every fixed neighborhood of the representative origin.

Fix m throughout the next steps. Let Q_R be the mass of endpoint pairs (u, v) with $v_2 - u_2 \geq R$ under

$$(\alpha_- K_{H_m}^{\Omega_m}) \otimes (\alpha_+ K_{H_m}^{\Omega_m}).$$

Every starting pair already has signed gap at least R . Its outward order event is therefore contained in the event just tested. Finite entropy transfers the probability-one event in (33) to Ω_m , simultaneously for the countably many starting pairs and integer heights, so it applies at the random height H_m . That estimate and Jensen's inequality yield

$$\begin{aligned} \log Q_R &\geq -A_{\rightarrow} \log(1 + H_m) - Z_R, \\ Z_R &= \iint \mathcal{Z}((0, x), (0, y); \Omega_m) \alpha_-(dx) \alpha_+(dy). \end{aligned} \tag{46}$$

Write $\nu_{\pm} = \alpha_{\pm} K_{H_m}^{\Omega_m}$. There is an integer j with $\nu_-(u_2 \leq j) \geq Q_R/2$ and $\nu_+(v_2 \geq j + R) \geq Q_R/2$. Choose j minimal for the first inequality, so $\nu_-(u_2 < j) < Q_R/2$. If the second inequality failed, split ordered pairs into $u_2 < j$ and $u_2 \geq j$. The latter condition forces $v_2 \geq j + R$. Each part would then have mass strictly less than $Q_R/2$, since both endpoint measures have mass at most one, contrary to the definition of Q_R .

Each initial tail has mass at least $T_R(w_0)$. Applying (43) to the two tails at this cut gives

$$\log T_R(w_m) - \log T_R(w_0) \geq C_m - A_{\rightarrow} \log(1 + H_m) - Z_R - \log 2. \tag{47}$$

This is the decisive inequality: normalization contributes C_m , whereas preserving the order of the two tails costs only a logarithm of the traversed height and the local environmental remainder.

We next remove the environmental bias from that remainder. We claim

$$\limsup_{R \rightarrow \infty} EZ_R \leq M_{\mathcal{Z}} := \sup_{x, y} E_P \mathcal{Z}(x, y) < \infty. \tag{48}$$

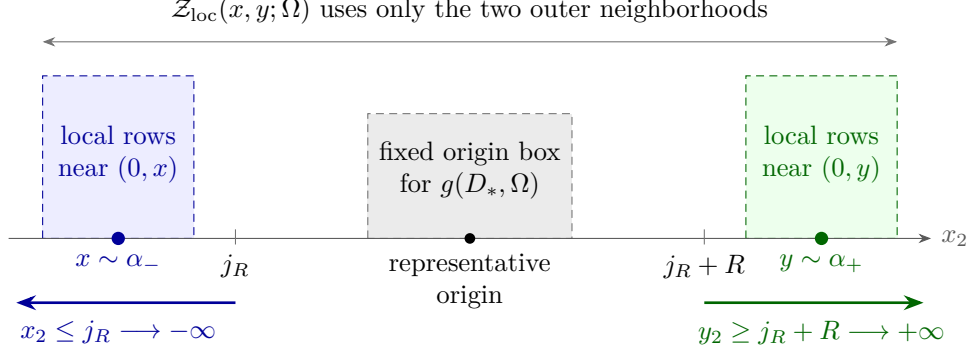


Figure 3: The finite-row approximations in the proof of Equation (48). At a fixed episode range, the bounded density approximation uses a fixed box around the representative origin. The local remainder uses neighborhoods of the two sampled tail sites. These neighborhoods leave the origin box as the tail parameter increases, so the approximating factors are conditionally independent under the product reference law. Only the second coordinate is shown.

Write $f_m(D, \Omega)$ for the density relative to (45). Under that product reference, additionally sample x, y from $\alpha_- \otimes \alpha_+$ conditionally on D alone. Then EZ_R is the reference expectation of $f_m(D, \Omega)\mathcal{Z}((0, x), (0, y); \Omega)$.

The exponential moment in (33) is uniform in the sampled pair. Consequently the density and the remainder can both be truncated with errors uniform in R . To see the only delicate part, on $\{f_m > B\}$ Young's inequality gives, for sufficiently small $\lambda > 0$,

$$\lambda f_m \mathcal{Z} \leq f_m \log f_m + e^{\lambda \mathcal{Z}}.$$

The entropy term has vanishing tail. The reference probability of $\{f_m > B\}$ is at most $1/B$, and Cauchy–Schwarz bounds the exponential term on that event by $O(B^{-1/2})$. Once f_m is bounded, truncating $\mathcal{Z}$ is justified directly by its exponential moment.

Approximate the bounded density by a bounded nonnegative function $g(D, \Omega)$ using only a fixed finite box of field rows around the origin, allowing arbitrary dependence on D . With L^1 error at most ε , its reference integral is at most $1 + \varepsilon$. Approximate the capped remainder, uniformly in x, y , by a bounded function using fixed-radius upper-row neighborhoods of those two sites. This follows from the local approximation in (33); boundedness converts convergence in probability to convergence in L^1 . As R increases, the two tail supports and their neighborhoods leave the fixed origin box. Under the product reference, conditional on D, x, y , the two approximating factors then use disjoint iid row sets. Their product expectation is bounded by $(M_{\mathcal{Z}} + \varepsilon)E_{P_{\dagger}}g(D, \cdot)$. The threshold in R may depend on D ; boundedness permits integration by reverse Fatou. Sending the approximation errors to zero and removing the truncations proves (48). This entire step keeps m fixed, so its approximation boxes need not be uniform in m .

For fixed R, m , the right side of (47) is integrable. The entropy inequality and the same exponential moment show that $EZ_R < \infty$. Let $X = \log T_R(w_0)$ and $Y = \log T_R(w_m)$. Translation invariance of T_R and stationarity make these finite nonpositive variables identically distributed. Their difference has integrable negative part by (47). Clipping both variables below at $-B$ gives an integrable zero-mean difference; its positive and negative parts increase to the corresponding parts of $Y - X$. Monotone convergence therefore shows that $Y - X$ is integrable and $E(Y - X) = 0$. Taking expectations in (47), and now sending R to infinity, yields

$$mEc_0 \leq A_{\rightarrow} E \log(1 + H_m) + M_{\mathcal{Z}} + \log 2.$$

Finally let m increase. With $Z_i = \log(1 + h_i)$,

$$\log(1 + H_m) \leq \log m + \max_{0 \leq i < m} Z_i, \quad E \max_{i < m} Z_i \leq T + mE[Z_0 \mathbf{1}_{\{Z_0 > T\}}].$$

Divide by m , send m to infinity and then T to infinity. The assumed logarithmic moment gives $E \log(1 + H_m) = o(m)$, hence $Ec_0 \leq 0$. Nonnegativity of c_0 proves the proposition. $\square$

Completion of the proof of Theorem 1.1. Apply Proposition 8.6 to $\mathbb{L}_*$. The profile construction supplies its update and moment hypotheses, and the marked field argument above supplies the finite entropy hypothesis. But on the shift-invariant event $\mathcal{E}$, the stationary ergodic theorem gives $E_{\mathbb{L}_*} c_0 > \zeta$. This contradicts the proposition and excludes the bad-event sequence Q_N . Thus (5) holds. Proposition 2.5 gives finite mean regeneration duration and an almost-sure deterministic vector velocity with positive first coordinate. This vector lies on the deterministic ray of Corollary 2.3, whose projection on the original transient direction ℓ is positive. Undoing the signed permutation of the coordinate axes therefore gives $X_n/n \rightarrow v$ with $v \cdot \ell > 0$, as asserted in Theorem 1.1. $\square$

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